THE HOPKINS–MOREL ISOMORPHISM IN POSITIVE CHARACTERISTIC

Abstract. Let k be a field of characteristic $p > 0$. We prove that the canonical map $\mathrm{MGL}/(a_1, a_2, \ldots) \to H\mathbb{Z}$, where the a_i are homogeneous polynomial generators of the Lazard ring, is an equivalence without inverting p. We also obtain the corresponding equivalence over $\operatorname{Spec}\mathbb{Z}_p$ for Spitzweck's motivic cohomology spectrum. After p-localization, we show in each positive degree that the image of weight-zero cobordism homotopy in the Lazard quotient has bounded exponent and is p-divisible, hence vanishes. The exponent bound combines integral Riemann–Roch with a Hilbert scheme approximation; divisibility follows by linearizing a geometric power congruence arising from a deformation of C_p to α_p. A filtration by Lazard weight and the Geisser–Levine theorem then yield the comparison.

Introduction

The Hopkins–Morel isomorphism describes motivic cohomology in terms of algebraic cobordism: $H\mathbb{Z}$ is obtained from MGL by killing the generators of the Lazard ring. Let $L = \mathbb{Z}[a_1, a_2, \ldots]$ be the Lazard ring, with a_i in bidegree $(2i, i)$. Since motivic cohomology carries the additive formal group law, its orientation induces a canonical comparison map

$$\Phi_k \colon \mathrm{MGL}/(a_1, a_2, \ldots) \longrightarrow H\mathbb{Z} \tag{0.1}$$

in $\mathrm{SH}(k)$. Here the quotient is the filtered colimit of the successive cofibres of multiplication by the a_i, taken in MGL-modules. Hoyois proved that (0.1) is an equivalence after inverting the characteristic exponent of k [H, Theorem 7.12 and Remark 7.15]. In this paper we prove that it is an equivalence without inverting the characteristic.

Theorem A. *Let p be a prime.*

(i) *For every field k of characteristic p, the map $\Phi_k \colon \mathrm{MGL}/(a_1, a_2, \ldots) \to H\mathbb{Z}$ is an equivalence in $\mathrm{SH}(k)$.*

(ii) *Let $M\mathbb{Z}$ be Spitzweck's motivic cohomology spectrum over $\operatorname{Spec}\mathbb{Z}_p$ [S]. Its additive orientation induces an equivalence*

$$\Phi_{\mathbb{Z}_p} \colon \mathrm{MGL}/(a_1, a_2, \ldots) \longrightarrow M\mathbb{Z} \quad \text{in } \mathrm{SH}(\operatorname{Spec}\mathbb{Z}_p).$$

For $E \in \mathrm{SH}(S)$ write $E^\wedge_p = \lim_r E/p^r$, where $E/p^r = \operatorname{cofib}(p^r \colon E \to E)$.

Corollary B. *The maps of Theorem A induce equivalences*

$$(\mathrm{MGL}/(a_1, a_2, \ldots))^\wedge_p \simeq (H\mathbb{Z})^\wedge_p \quad \text{over } k, \qquad (\mathrm{MGL}/(a_1, a_2, \ldots))^\wedge_p \simeq M\mathbb{Z}^\wedge_p \quad \text{over } \operatorname{Spec}\mathbb{Z}_p.$$

In this corollary the Lazard quotient is formed before completion.

The idea of the proof. The Lazard ring carries the universal formal group law, and the augmentation $L \to \mathbb{Z}$ that kills the a_i corresponds to the additive one. By Hoyois's theorem, the only question is whether the comparison map has a p-primary error, so we may work p-locally. All three spectra MGL, $\mathrm{MGL}/(a_1, a_2, \ldots)$ and $H\mathbb{Z}$ are compatible with base change from $\mathbb{F}_p$ to k, so we may moreover take $k = \mathbb{F}_p$ (Theorem 6.1). Write $M = \mathrm{MGL}_{(p)}$ and $\Lambda_{(p)} = (\mathrm{MGL}/(a_1, a_2, \ldots))_{(p)}$. Bachmann's commutative realization of L [B] makes $\Lambda_{(p)}$ a commutative M-algebra, which we may think of as cobordism specialized to the additive formal group law. Let $\rho \colon M \to \Lambda_{(p)}$ be its structure map, and let $\Phi \colon \Lambda_{(p)} \to H\mathbb{Z}_{(p)}$ be the p-localization of $\Phi_{\mathbb{F}_p}$. We show that Φ is an equivalence by combining three principles.

A first obstruction comes from cobordism. Suppose that Φ is not an equivalence, and choose s minimal such that the fibre F of Φ does not lie in $\mathrm{SH}(\mathbb{F}_p)_{\geq s+1}$ for the homotopy t-structure, that is, $F \in \mathrm{SH}(\mathbb{F}_p)_{\geq s}$ and $h_s(F) \neq 0$ (homotopy modules, see Conventions).

The nonvanishing of $h_s(F)$ is detected by a group of field values. The spectrum F is effective, because $\Lambda_{(p)}$ and $H\mathbb{Z}_{(p)}$ are. Hence Proposition 5.3 applies: weight zero controls all weights, and some group $\pi_{s,0}F(K)$ at a finitely generated field $K/\mathbb{F}_p$ is nonzero. This group is described by the following facts.

- The spectra $\Lambda_{(p)}$ and $H\mathbb{Z}_{(p)}$ are connective.
- $\pi_{j,0}H\mathbb{Z}_{(p)}(K) = 0$ for $j \neq 0$.
- Φ is an isomorphism on $\pi_{0,0}$.

Together they give $s \geq 1$ and $\pi_{s,0}F(K) = \pi_{s,0}\Lambda_{(p)}(K)$. By Hoyois's theorem this is a p-primary torsion group.

To relate it to cobordism, we filter M by Lazard weight. The graded pieces are copies $\Sigma^{2i,i}\Lambda_{(p)}$ of $\Lambda_{(p)}$ with $i > 0$, and $\underline{\pi}_{s,0}(\Sigma^{2i,i}\Lambda_{(p)}) = h_{s-i}(\Lambda_{(p)})_i$ lies in degree $s - i < s$: a Tate twist moves homotopy into higher connectivity. In these degrees Φ is an equivalence by the minimality of s. By the theorem of Geisser and Levine, $H\mathbb{Z}/p$ has no higher homotopy. Using these two facts, a Bockstein argument shows that every p-primary class in $\pi_{s,0}\Lambda_{(p)}(K)$ lifts to $\pi_{s,0}M(K)$ (Lemma 5.10). Hence it suffices to show that

$$\rho_*\colon \pi_{s,0}M(K) \longrightarrow \pi_{s,0}\Lambda_{(p)}(K) \quad \text{is zero for } s > 0.$$

This is a statement about cobordism, and the positive homotopy of cobordism is represented by families of finite syntomic algebras over spheres [BEHKSWY], so the statement can be attacked geometrically. Let $I_s(K)$ be the image of ρ_*. We show that $I_s(K)$ is p-divisible and has bounded exponent.

Rigidity: the image is p-divisible. The first input is a geometric form of Fermat's little theorem $d^p \equiv d \pmod p$. For a finite syntomic morphism $f\colon W \to X$ we put $u = f_*^M 1$ and prove the necklace formula

$$\rho(u^p - u) = p\,\rho(b) \qquad \text{for some } b \in M^{0,0}(X). \tag{0.2}$$

Here b is a cobordism lift of the class of the stack of non-constant necklaces in W. When X is a point and W consists of d rational points, the formula says that $d^p - d$ is p times the number of non-constant necklaces of length p on d letters.

To deduce divisibility, we use that a polynomial operation acts on positive homotopy by its derivative, because reduced products of classes on a sphere vanish (Lemma 5.1). The polynomial $P(t) = t^p - t$ has derivative $P'(t) = pt^{p-1} - 1$, which is a unit modulo p; in other words, the equation $t^p = t$ is étale. We represent a class $x \in \pi_{s,0}M(K)$ by a family of degree d over a sphere, so that $u = d + x$, and we write $b = b_0 + y$ with b_0 constant and y in positive homotopy. Since reduced products vanish, we have $\rho(u^p - u) = d^p - d + (pd^{p-1} - 1)\rho(x)$. Comparing with (0.2), we find $d^p - d = pb_0$ and

$$(pd^{p-1} - 1)\,\rho(x) = p\,\rho(y).$$

Because b is a class in cobordism, and not merely in $\Lambda_{(p)}$, the element y lies in $\pi_{s,0}M(K)$, so $\rho(y)$ lies in $I_s(K)$. Since $pd^{p-1} - 1$ is a unit in $\mathbb{Z}_{(p)}$, every element $\rho(x)$ of $I_s(K)$ is p times an element of $I_s(K)$. In other words, $I_s(K) = pI_s(K)$.

Finite dimensionality: the image has bounded exponent. The second input is a bound on the order of elements of $I_s(K)$ that does not depend on the degree of the families. For a family $W \subset \mathbb{P}^n_X$ of degree d, an integral Riemann–Roch identity gives $(n-1)!\,(f_*^{\Lambda_{(p)}}1 - d) = 0$ in $\Lambda^{0,0}_{(p)}(X)$. To make n depend only on s, we use that a tuple of functions generates a finite algebra exactly when it generates every quotient of length two. Failing to generate a fixed length-two quotient is $s+1$ linear conditions on an $(s+1)$-tuple, and the incidence space of pairs (algebra, length-two quotient) has the dimension of the space of algebras of degree d. Hence the non-generating $(s+1)$-tuples have codimension at least $s+1$. Removing a closed subset of codimension at least $s+1$ loses no classes in π_s (Corollary 2.6). Hence every class in $\pi_{s,0}M(K)$ comes from a family embedded in $\mathbb{A}^{s+1} \subset \mathbb{P}^{s+1}$, that is, from a Hilbert scheme

of points of $\mathbb{A}^{s+1}$. Taking $n = s+1$ in the Riemann–Roch identity, we obtain $s!\, I_s(K) = 0$. A p-divisible $\mathbb{Z}_{(p)}$-module of bounded exponent is zero, so $I_s(K) = 0$.

The necklace formula (0.2) is the counting proof of $d^p = d + p \cdot \#\{\text{non-constant necklaces}\}$, transported into geometry. We deform W^p to the normal cone of the diagonal. In this deformation the diagonal contributes f_*1, and the rest carries an action of a group of order p, which contributes $p\, b$ once the action is free on a proper model. The natural proper model of the complement of the diagonal is the blowup $\mathrm{Bl}_\Delta W^p$ of W^p along the diagonal.

In characteristic p the cyclic group does not act freely on this model, and we replace it by the group scheme α_p. A free action on a proper model exists only after we deform C_p to α_p. This deformation turns p-tuples into jets and the blowup $\mathrm{Bl}_\Delta W^p$ into a weighted blowup, which is the proper model on which the quotient is formed. The factor p comes from the degree of the free C_p- (resp. α_p-) quotient.

Tameness is what makes the classes of these quotients computable. The pieces of the weighted blowup are tame stacks, that is, stacks whose stabilizers are linearly reductive. Tameness allows us to compute their classes on finite covers of degree prime to p.

Once the theorem is proved, the necklace formula identifies b in every additive theory. It is $(d^p - d)/p$, the number of non-constant necklaces of length p on d letters, also for inseparable or nonreduced families (Corollary 7.2).

Organization. Section 1 sets up the commutative model $\Lambda_{(p)}$, the map Φ and the rules for computing fundamental classes. Section 2 proves the bounded exponent $s!\, I_s(K) = 0$. Section 3 shows how to compute classes of tame stacks on finite covers of degree prime to p, and Section 4 uses this to prove the necklace formula (0.2). Section 5 deduces $I_s(K) = 0$ and, with the lifting lemma, the theorem over $\mathbb{F}_p$. Section 6 passes to arbitrary fields, to $\operatorname{Spec}\mathbb{Z}_p$, and to derived completion, and Section 7 records what the theorem says about additive theories.

Conventions. We use the Nisnevich motivic stable category $\mathrm{SH}(S)$ and its S^1-stable version $\mathrm{SH}^{S^1}(S)$. For a motivic spectrum E,

$$E^{a,b}(X) = [\Sigma^\infty X_+, \Sigma^{a,b} E], \qquad \pi_{a,b} E(X) = [\Sigma^{a,b}\Sigma^\infty X_+, E],$$

where $\Sigma^{2,1}$ is suspension by $\mathbb{P}^1$. We write L_{mot} for Nisnevich and $\mathbb{A}^1$ localization, and take values at essentially smooth fields by filtered colimits over smooth models. The homotopy modules are indexed by

$$h_i(E)_m = \underline{\pi}_{i-m,-m} E;$$

in particular the weight-zero sheaf is $h_i(E)_0 = \underline{\pi}_{i,0} E$. Derived schemes and stacks are modelled on animated commutative rings, t_0 denotes classical truncation, and dimensions refer to t_0. The notions of derived geometry that we need, quasi-smooth morphisms and their fundamental classes, are recalled in Section 1. The projective bundle $\mathbb{P}(V)$ parametrizes lines, with tautological subbundle $\mathcal{O}(-1)$. Except in Section 6, the base is $\mathbb{F}_p$.

1. The Lazard quotient, the comparison map and fundamental classes

This section sets up the objects that the rest of the paper computes with: a model of the Lazard quotient as a commutative ring spectrum (Section 1.1), the comparison map Φ (Section 1.2), fundamental classes of quasi-smooth maps (Section 1.3), and derived Weil restriction (Section 1.4).

1.1. **A commutative model of the Lazard quotient.**

The problem. The spectrum $\mathrm{MGL}/(a_1, a_2, \ldots)$ of the introduction is defined as a module. One forms $\mathrm{MGL}/(a_1)$ as the cofibre of multiplication by a_1 on MGL, then the cofibre of a_2 on the result, and so on, in MGL-modules, and takes the colimit. A cofibre of a map of modules has no induced multiplication, so this definition does not make the quotient a ring spectrum. We need more than a module for three reasons.

(i) The necklace formula (0.2) is a statement about p-th powers of classes in the cohomology of the quotient.
(ii) The trace computations of Sections 3 and 4 use that the quotient is oriented with the additive formal group law, so that $c_1(L \otimes L') = c_1(L) + c_1(L')$.
(iii) The argument of Section 5 uses a filtration of $\mathrm{MGL}_{(p)}$ whose graded pieces are shifted copies of the quotient.

Item (i) needs ρ to be a map of ring spectra; items (ii) and (iii) use the base-change description of the quotient constructed below. In topology $MU/(x_1, x_2, \ldots) \simeq H\mathbb{Z}$ is a ring spectrum after the fact. Here the ring structure is needed in the proof of the comparison, so it has to be constructed directly.

The idea. In algebra the quotient of L by all the a_i is a base change: if $\epsilon\colon L \to \mathbb{Z}$ is the augmentation that kills every a_i, then $L/(a_1, a_2, \ldots) = L \otimes_L \mathbb{Z}$, and the Koszul complex shows that $-\otimes_L \mathbb{Z}$ kills $a_1, a_2, \ldots$ one at a time. So it suffices to realize three things in $\mathrm{SH}(\mathbb{F}_p)_{(p)}$: the polynomial ring $L_{(p)}$ as a commutative algebra $\mathcal{A}$, with a generator of degree i in bidegree $(2i, i)$; the classifying map of the formal group law as a map of commutative algebras $\mathcal{A} \to \mathrm{MGL}_{(p)}$; and the augmentation ϵ. The base change $\mathrm{MGL}_{(p)} \otimes_{\mathcal{A}}$ (realization of $\mathbb{Z}_{(p)}$) is then a commutative algebra, and the Koszul complex identifies its underlying module with the iterated quotient.

Why the grading must be changed. The natural way to realize a $\mathbb{Z}$-graded object is to send its part in degree n to its tensor product with $(\Sigma^{2,1}\mathbf{1})^{\otimes n}$. To make this functor symmetric monoidal one has to match the switch maps. In $\mathbb{Z}$-graded objects the switch of two factors of degree 1 is the identity, whereas the switch of the two factors of $\Sigma^{2,1}\mathbf{1} \otimes \Sigma^{2,1}\mathbf{1}$ is the class $\langle -1\rangle$ (the sign -1 of $S^1 \wedge S^1$ times $\epsilon = -\langle -1\rangle$ on $\mathbb{G}_m \wedge \mathbb{G}_m$), and a symmetric monoidal functor would need a coherent trivialization of it, which is extra structure that one does not want to construct. Bachmann's remedy is to grade by the free grouplike $\mathbb{E}_\infty$-space $\mathbb{S}$ on one generator, whose set of components is $\mathbb{Z}$. An $\mathbb{S}$-graded spectrum is a functor from $\mathbb{S}$ to spectra, and since $\mathbb{S}$ is free, a symmetric monoidal functor out of $\mathbb{S}$-graded spectra is determined by an invertible object, with no condition on its switch map. We write U for the functor that regards a $\mathbb{Z}$-graded object as an $\mathbb{S}$-graded one, by restriction along $\mathbb{S} \to \pi_0\mathbb{S} = \mathbb{Z}$. With respect to Day convolution, restriction has a lax symmetric monoidal structure, and therefore takes commutative algebras and their modules to commutative algebras and their modules.

Theorem 1.1 ([B, §6.5 and Corollary 6.28])**.** *There is a colimit-preserving symmetric monoidal functor $e_{\mathbb{S}}$ from $\mathbb{S}$-graded spectra to $\mathrm{SH}(\mathbb{F}_p)_{(p)}$ that sends the generator of $\mathbb{S}$ to $\Sigma^{2,1}\mathbf{1}$. There is a map of commutative algebras*

$$e_{\mathbb{S}}U(L_{(p)}) \longrightarrow \mathrm{MGL}_{(p)}$$

which realizes the classifying map $L_{(p)} \to \pi_{2*,*}\mathrm{MGL}_{(p)}$ of the formal group law of $\mathrm{MGL}_{(p)}$; its homotopy class is canonical.

Bachmann uses the same construction to give the Brown–Peterson spectrum an $\mathbb{E}_\infty$-structure, as $\mathrm{MGL}_{(p)} \otimes_{e_{\mathbb{S}}U(L_{(p)})} e_{\mathbb{S}}U(BP_*)$ [B, Example 6.29]. We use it with $\mathbb{Z}_{(p)}$ in place of BP_*.

Definition 1.2. Let $\mathcal{A} = e_{\mathbb{S}}U(L_{(p)})$, choose a map $\mathcal{A} \to \mathrm{MGL}_{(p)}$ as in Theorem 1.1, and let $\mathcal{A} \to e_{\mathbb{S}}U(\mathbb{Z}_{(p)})$ be induced by the augmentation $\epsilon\colon L_{(p)} \to \mathbb{Z}_{(p)}$. Put

$$\Lambda_{(p)} = \mathrm{MGL}_{(p)} \otimes_{\mathcal{A}} e_{\mathbb{S}}U(\mathbb{Z}_{(p)}), \tag{1.1}$$

and let $\rho\colon M := \mathrm{MGL}_{(p)} \to \Lambda_{(p)}$ be the unit map.

We view graded rings and complexes of graded modules as Eilenberg–Mac Lane objects in graded spectra. Restriction U is exact and preserves colimits, since these are computed pointwise. We shall also need its compatibility with grading shifts, including the module structures. For a graded $L_{(p)}$-module N, write $N(w)_j = N_{j-w}$. Let δ_w be the free $\mathbb{S}$-graded sphere represented by the w-fold translate of the chosen generator; its value at $t \in \mathbb{S}$ is $\Sigma^\infty_+ \operatorname{Map}_{\mathbb{S}}(w, t)$. Since $\mathbb{S}$ is grouplike, the Day convolution Yoneda formula gives

$$(\delta_w \otimes_{\mathrm{Day}} F)(t) \simeq F(t - w).$$

Restriction along $\mathbb{S} \to \mathbb{Z}$ commutes with this translation. Consequently

$$U(N(w)) \simeq \delta_w \otimes_{\mathrm{Day}} U(N), \qquad e_{\mathbb{S}} U(N(w)) \simeq \Sigma^{2w,w} e_{\mathbb{S}} U(N). \tag{1.2}$$

The first equivalence is one of $U(L_{(p)})$-modules: both actions are induced by addition in the grading space and the action on N. The second is therefore an equivalence of $\mathcal{A}$-modules, using the symmetric monoidal functor $e_{\mathbb{S}}$ and $e_{\mathbb{S}}(\delta_w) = \Sigma^{2w,w}\mathbf{1}$. Under these identifications, the map $a_i\colon N(i) \to N$ is the module action of the map $\delta_i \to U(L_{(p)})$ representing a_i. After realization and base change along $\mathcal{A} \to M$, it is multiplication by the corresponding class in $\pi_{2i,i}M$, by Theorem 1.1.

Proposition 1.3. (i) *$\Lambda_{(p)}$ is a commutative algebra over $\mathrm{MGL}_{(p)}$, and ρ is a map of commutative algebras.*
(ii) *The underlying $\mathrm{MGL}_{(p)}$-module of $\Lambda_{(p)}$ is the p-localization of $\mathrm{MGL}/(a_1, a_2, \ldots)$.*
(iii) *The orientation of $\Lambda_{(p)}$ obtained from that of MGL through ρ has the additive formal group law.*

Proof. Part (i) holds because (1.1) is a base change of commutative algebras. For (ii), consider the functor $N \mapsto \mathrm{MGL}_{(p)} \otimes_{\mathcal{A}} e_{\mathbb{S}} U(N)$ from graded $L_{(p)}$-modules to $\mathrm{MGL}_{(p)}$-modules. The lax structure of U provides the module structure used here. This functor is exact and preserves colimits: U and $e_{\mathbb{S}}$ have these properties, and derived tensor product preserves colimits and cofibre sequences. It sends $L_{(p)}$ to $\mathrm{MGL}_{(p)}$. By (1.2) and its compatibility with the module action, it sends a shift by weight w to $\Sigma^{2w,w}$ and multiplication by a_i to multiplication by the image of a_i in $\pi_{2i,i}\mathrm{MGL}_{(p)}$. The $L_{(p)}$-module $\mathbb{Z}_{(p)}$ is the colimit of the quotients $L_{(p)}/(a_1, \ldots, a_N)$, and $L_{(p)}/(a_1, \ldots, a_{N+1})$ is the cofibre of multiplication by a_{N+1} on the weight-$(N+1)$ shift of $L_{(p)}/(a_1, \ldots, a_N)$, because the a_i form a regular sequence. Applying the functor gives exactly the iterated cofibres defining $\mathrm{MGL}_{(p)}/(a_1, a_2, \ldots)$.

For (iii), the coefficients of the formal group law of $\Lambda_{(p)}$ are the images of those of $\mathrm{MGL}_{(p)}$, that is, the images of $L_{(p)}$ under $\mathcal{A} \to \mathrm{MGL}_{(p)} \to \Lambda_{(p)}$. In the base change (1.1) this composite agrees with $\mathcal{A} \to e_{\mathbb{S}} U(\mathbb{Z}_{(p)}) \to \Lambda_{(p)}$, which factors through ϵ. So every a_i maps to 0, and the formal group law is additive. □

1.2. **The comparison map.** Theorem A is about one specific map, the map Φ of (0.1), which is a map of modules and need not respect the ring structure of Proposition 1.3. We construct it over $\mathbb{F}_p$ after p-localization. Write $\Lambda_N = \mathrm{MGL}_{(p)}/(a_1, \ldots, a_N)$; here Λ_N is already p-local.

Lemma 1.4. *There is a map of $\mathrm{MGL}_{(p)}$-modules $\Phi\colon \Lambda_{(p)} \to H\mathbb{Z}_{(p)}$ extending the unit $\mathrm{MGL}_{(p)} \to H\mathbb{Z}_{(p)}$, and it is unique up to a contractible space of choices. It is the p-localization of (0.1).*

Proof. The point is that $H\mathbb{Z}_{(p)}$ admits no maps from positive Tate twists of $\mathrm{MGL}_{(p)}$-cells:

$$\operatorname{map}_{\mathrm{MGL}_{(p)}}(\Sigma^{2i,i}\Lambda_N, H\mathbb{Z}_{(p)}) = 0 \qquad (i > 0). \tag{1.3}$$

Indeed, Λ_N is a finite iterated cofibre of cells $\Sigma^{2w,w}\mathrm{MGL}_{(p)}[q]$ with $w, q \geq 0$, and the mapping spectrum from such a cell twisted by $\Sigma^{2i,i}$ is, up to shift, motivic cohomology of $\operatorname{Spec}\mathbb{F}_p$ in weight $-(w+i) < 0$, which vanishes. Since Λ_{N+1} is the cofibre of $a_{N+1}\colon \Sigma^{2N+2,N+1}\Lambda_N \to \Lambda_N$,

(1.3) shows that a map $\Lambda_N \to H\mathbb{Z}_{(p)}$ extends to Λ_{N+1} uniquely up to a contractible choice. By induction and passage to the colimit we obtain Φ, and the same uniqueness shows that it agrees with the p-localization of (0.1). □

1.3. **Fundamental classes.** For a smooth proper morphism $f\colon W \to X$ of smooth schemes and an oriented theory E, the Gysin pushforward gives a class $f_*1 \in E^{-2v,-v}(X)$, where v is the relative dimension of f. It is the fundamental class of W over X. The whole proof is a computation with such classes $[W \to X]$, but for maps $W \to X$ that are not smooth: W is a finite syntomic family, a power W^p, or one of the spaces built from them in Section 4. These spaces are in general neither smooth nor classical schemes. What they have in common is that they are quasi-smooth derived schemes over X, the derived analogue of local complete intersections, and quasi-smooth morphisms have fundamental classes. We recall the definition and the facts we use.

Definition 1.5. A morphism $f\colon W \to X$ of derived schemes is *quasi-smooth* if it is locally of finite presentation and its cotangent complex $\mathbb{L}_{W/X}$ is perfect of Tor-amplitude in $[-1, 0]$; its *virtual relative dimension* v is the rank of $\mathbb{L}_{W/X}$. It is *smoothable* if it factors as a quasi-smooth closed immersion $W \to P$ followed by a smooth morphism $P \to X$.

The condition on $\mathbb{L}_{W/X}$ says that, locally, W is cut out of a smooth X-scheme by finitely many equations, and that the virtual dimension is the number of variables minus the number of equations. Two examples cover all our uses.

(i) Let V be a vector bundle of rank r on a smooth scheme X, and let s be a section. The derived zero locus $Z(s) = X \times_{s,V,0} X$ is a closed derived subscheme of X, quasi-smooth of virtual dimension $-r$ over X, whatever the dimension of its underlying classical scheme.
(ii) A finite syntomic morphism is quasi-smooth of virtual dimension 0.

Borel–Moore homology. To compute with fundamental classes of non-smooth spaces we need a place for them to live. Fix a scheme X of finite type over $\mathbb{F}_p$ (in the final applications X is smooth and affine, but we also need derived bases such as the spaces of Section 4; everything below holds over any derived scheme X of finite type, since the six operations exist there [K, §2, Appendix A]) and an oriented MGL-algebra E. For a derived scheme $\pi\colon W \to X$ of finite type, Khan defines Borel–Moore homology and cohomology [K, Definition 2.1]:

$$H_s^{\mathrm{BM}}(W/X, E(r)) = [\mathbf{1}_X(r)[s], \pi_*\pi^!E], \qquad E^{s,r}(W) = [\mathbf{1}_X, \pi_*\pi^*E(r)[s]].$$

They depend only on the underlying classical scheme t_0W, because the six operations do. If W is smooth over X of dimension v, purity identifies $\pi^!E$ with $\pi^*E(v)[2v]$, and Borel–Moore homology with cohomology shifted by $(2v, v)$; for $W = X$ it is $E^{*,*}(X)$. In general the two theories carry the following operations [K, §§2.2–2.3], which satisfy the usual compatibilities of a bivariant theory.

- Proper pushforward g_* on Borel–Moore homology, for proper $g\colon W \to W'$ over X.
- Base change h^* along $h\colon X' \to X$, using derived fibre products.
- Cap product of a cohomology class of W with a Borel–Moore class of W.
- Euler classes $c_r(V) \in E^{2r,r}(W)$ of vector bundles on W.

When W is smooth over X these are the familiar operations on E-cohomology.

Theorem 1.6 ([K, Variant 3.11, Theorems 3.12 and 3.13, Corollary 3.17])**.** *Every smoothable quasi-smooth morphism $\pi\colon W \to X$ of virtual dimension v has a fundamental class*

$$[W/X] \in H_{2v}^{\mathrm{BM}}(W/X, E(v)),$$

and these classes satisfy the following.

(i) Composition. *If $W \to W' \to X$ are smoothable quasi-smooth, then $[W/X] = [W/W'] \circ [W'/X]$, where $\circ$ is the composition product of bivariant classes* [K, 2.2.5]. *Fundamental*

classes are also compatible with external products and natural in maps of oriented ring spectra $E \to E'$ [K, §2.3], [DJK, §3].

(ii) Base change. *For any $h\colon X' \to X$, $h^*[W/X] = [W'/X']$ with $W' = W \times_X X'$ the derived fibre product.*

(iii) Self-intersection. *For a quasi-smooth closed immersion $i\colon W \to P$ of virtual codimension r with conormal bundle N, $i^*i_*[W/P] = c_r(N^\vee)$ in $E^{2r,r}(W)$.*

We shall use (iii) through the following consequence. Let V be a vector bundle of rank r on P, s a section and $i\colon Z(s) \to P$ its derived zero locus. Then $Z(s)$ is the base change of the zero section $0\colon P \to V$ along s, so $i_*[Z(s)/P] = s^*0_*1$ by (ii) and proper base change. As s and 0 are both sections of the $\mathbb{A}^1$-equivalence $V \to P$, $s^* = 0^*$, and (iii) for the zero section, whose conormal bundle is $V^\vee$, gives

$$i_*[Z(s)/P] = c_r(V) \quad \text{in } E^{2r,r}(P). \tag{1.4}$$

If X is smooth and π is proper, we push $[W/X]$ forward to X and obtain a class

$$[W \to X]_E = \pi_*[W/X] \in H^{\mathrm{BM}}_{2v}(X/X, E(v)) = E^{-2v,-v}(X).$$

This is the fundamental class of W over X; for $W \to X$ smooth and proper it is the Gysin pushforward π_*1. More generally, for a proper smoothable quasi-smooth $h\colon P \to W$ of virtual dimension 0 over X, we call $h_*[P/W] \in H^{\mathrm{BM}}_0(W/W, E) = E^{0,0}(W)$ the *trace* of h and write it h_*1. We use $E = M = \mathrm{MGL}_{(p)}$ and $E = \Lambda_{(p)}$, and we write $[W \to X]$ when E is clear. By (i), a map of oriented ring spectra $E \to E'$ sends $[W \to X]_E$ to $[W \to X]_{E'}$. The following rules are all that we use.

Corollary 1.7. *Let X be a derived scheme of finite type, and let $W \to X$ and $W' \to X$ be proper smoothable quasi-smooth morphisms; write $[W \to X] \in E^{*,*}(X)$ for the pushforward of $[W/X]$.*

(i) Base change. *For a map $g\colon X' \to X$, $g^*[W \to X] = [W \times_X X' \to X']$, with derived fibre product.*

(ii) Products. $[W \to X] \cdot [W' \to X] = [W \times_X W' \to X]$.

(iii) Composition. *Let $h\colon P \to W$ be proper, smoothable and quasi-smooth of virtual dimension 0, with W not necessarily smooth. If its trace $h_*1 \in E^{0,0}(W)$ of Section 1.3 is an integer, $h_*1 = r \cdot 1$, then $[P \to X] = r[W \to X]$.*

(iv) Euler class. *Let $\pi\colon P \to X$ be proper, smoothable and quasi-smooth, let V be a vector bundle of rank r on P with a section s, and let $Z(s) \subset P$ be its derived zero locus. Then $[Z(s) \to X] = \pi_*(c_r(V) \cap [P/X])$. If $P \to X$ is smooth, this is $\pi_*c_r(V)$.*

Proof. (i) is Theorem 1.6(ii) followed by proper base change. For (ii), the class of $W \times W' \to X \times X$ is the external product of the two classes, and we pull it back along the diagonal using (i). For (iii), Theorem 1.6(i) gives $[P/X] = [P/W] \circ [W/X]$; pushing forward along h and using the projection formula for the composition product, $h_*[P/X] = h_*[P/W] \cap [W/X] = r\,[W/X]$, and we push forward to X. For (iv), Theorem 1.6(i) applied to $Z(s) \to P \to X$ and (1.4) show that $[Z(s)/X]$, pushed forward to P, is $c_r(V) \cap [P/X]$; we push forward to X; when P is smooth over X, $[P/X]$ is the orientation class and $c_r(V) \cap [P/X]$ corresponds to $c_r(V)$. □

Theorem 1.6 applies only to smoothable maps. All our spaces carry a line bundle, coming from their construction, that is ample on the underlying classical scheme. The following lemma shows that, over a smooth affine base, such a line bundle on a space proper over the base gives the factorization required for smoothability.

Lemma 1.8. *Let X be a smooth affine scheme, let P and S be derived schemes quasi-smooth over X, and let $g\colon P \to S$ be a quasi-smooth morphism, with S separated over X. Suppose that t_0P is proper over X and that P carries a line bundle L whose restriction to t_0P is ample. Then g factors as a quasi-smooth closed immersion $P \to \mathbb{P}^r_S = \mathbb{P}^r_X \times_X S$ followed by the projection.*

Proof. The idea is to embed t_0P into projective space by sections of a power of L, and to check that these sections lift to P. First, the homotopy sheaves $\pi_j\mathcal{O}_P$ are coherent and vanish for j large, since locally P is cut out of a smooth X-scheme by finitely many equations. By Serre vanishing, for m large we have $H^a(t_0P, \pi_j\mathcal{O}_P \otimes L^m) = 0$ for all $a > 0$ and all j. Hence the descent spectral sequence gives $\pi_0\Gamma(P, L^m) = \Gamma(t_0P, L^m)$. For m large there are finitely many sections of L^m on t_0P defining a closed immersion $t_0P \to \mathbb{P}^r_X$, and by the previous identification we may lift them to P.

The resulting map $j\colon P \to \mathbb{P}^r_X$ is a closed immersion, since its classical truncation is a closed immersion. Since S is separated over X, the graph of g is a closed immersion $P \to P \times_X S$. Composing with the base change of j gives a closed immersion $i = (j, g)\colon P \to \mathbb{P}^r_S$. It is locally of finite presentation, since the schemes involved are locally of finite presentation over X.

To show that i is quasi-smooth, use the cotangent triangle over S:

$$i^*\Omega_{\mathbb{P}^r_S/S} \longrightarrow \mathbb{L}_{P/S} \longrightarrow \mathbb{L}_{P/\mathbb{P}^r_S}.$$

The first term is a vector bundle in degree zero, and the second is perfect of Tor-amplitude $[-1, 0]$, by the quasi-smoothness of g. Thus $\mathbb{L}_{P/\mathbb{P}^r_S}$ is perfect of Tor-amplitude $[-1, 0]$. Moreover

$$H^0(\mathbb{L}_{P/\mathbb{P}^r_S}) = \Omega_{t_0P/t_0\mathbb{P}^r_S} = 0.$$

For a geometric point $x\colon \operatorname{Spec}\kappa(x) \to P$, connectivity gives $H^0(x^*\mathbb{L}_{P/\mathbb{P}^r_S}) = H^0(\mathbb{L}_{P/\mathbb{P}^r_S}) \otimes \kappa(x) = 0$. The Tor-amplitude bound then shows that this geometric fibre is concentrated in cohomological degree -1.

Consequently $\mathbb{L}_{P/\mathbb{P}^r_S}[-1]$ is connective and its geometric fibres are finite-dimensional vector spaces in degree zero. Locally lift a basis of such a fibre to sections of $H^0(\mathbb{L}_{P/\mathbb{P}^r_S}[-1])$. They define a map from a finite free $\mathcal{O}_P$-module to $\mathbb{L}_{P/\mathbb{P}^r_S}[-1]$ that is an equivalence at x. Its cofibre is perfect and has zero fibre at x, so derived Nakayama makes this map an equivalence in a neighbourhood of x. Thus $\mathbb{L}_{P/\mathbb{P}^r_S}[-1]$ is finite locally free, proving that i is quasi-smooth. The projection $\mathbb{P}^r_S \to S$ is smooth, as required. □

The traces in Sections 3 and 4 are traces in $\Lambda_{(p)}$ of zero loci in projective bundles, computed by Corollary 1.10 below. Since $\Lambda_{(p)}$ has the additive formal group law, first Chern classes are additive on tensor products of line bundles, and these computations reduce to the following form of the projective bundle formula.

Lemma 1.9. *Let E be an additively oriented* MGL*-algebra, S a derived scheme, V a vector bundle of rank $r+1$ on S, $\pi\colon \mathbb{P}(V) \to S$ and $h = c_1(\mathcal{O}(1))$. Then $\pi_*h^r = 1$ and $\pi_*h^i = 0$ for $0 \le i < r$.*

Proof. The idea is to compute the class of the diagonal of $\mathbb{P}(V)$ in two ways. Put $Q = \pi^*V/\mathcal{O}(-1)$, a bundle of rank r. On $\mathbb{P}(V) \times_S \mathbb{P}(V)$ the diagonal is the derived zero locus of the section of $\mathcal{O}(1)_1 \otimes Q_2$ given by $\mathcal{O}(-1)_1 \to V \to Q_2$; this can be checked on fibres over S, where the section is regular. Here and below, for a derived scheme S, $E^{*,*}(S)$ is the cohomology of Section 1.3 and the Euler class formula is (1.4). By the Euler class formula and additivity of c_1, we get $[\Delta] = \sum_{j=0}^r h_1^j\, c_{r-j}(Q_2)$. This is the only place where additivity enters: $c(\mathcal{O}(1)_1 \otimes Q_2)$ is computed from the Chern roots of Q_2 shifted by h_1, and for a general formal group law the identity fails (for $E = \mathrm{MGL}$ one has $\pi_*1 = [\mathbb{P}^r] \neq 0$). Pushing forward along the second projection gives $1 = \sum_{j=0}^r \pi^*(\pi_*h^j)\, c_{r-j}(Q)$. By the Whitney formula, $c_m(Q) = \sum_{i=0}^m \pi^*c_i(V)h^{m-i}$, so the right hand side is $\sum_j \pi^*(\pi_*h^j)h^{r-j}$ plus terms in which h has smaller exponent. We compare coefficients in the basis $1, h, \dots, h^r$ of the projective bundle formula, from h^r downward. Inductively, the coefficients of $h^r, \dots, h^1$ vanish and give $\pi_*h^i = 0$ for $0 \le i < r$. The coefficient of h^0 is then π_*h^r, and it equals 1. See also [D, Lemma 5.2.5]. □

Corollary 1.10. *Let E, S, V and π be as in Lemma 1.9, let $e_1, \dots, e_r$ be integers, $A_1, \dots, A_r$ line bundles on S, and let $Z \subset \mathbb{P}(V)$ be the derived zero locus of a section of $\bigoplus_i \mathcal{O}(e_i) \otimes \pi^* A_i$, with $E^{0,0}(S)$ and traces as in Section 1.3. If $Z \to S$ is finite, then $[Z \to S]_E = e_1 \cdots e_r$, and this holds after every derived base change $T \to S$.*

Finiteness is there so that $Z \to S$ is proper of virtual dimension 0, and $[Z \to S]_E$ is a trace in the sense of Section 1.3.

Proof. By the Euler class formula, $[Z \to S]_E = \pi_* \prod_i (e_i h + \pi^* c_1(A_i))$. By Lemma 1.9, only the term $e_1 \cdots e_r h^r$ survives. Since the derived zero locus is compatible with derived base change, the same argument applies over every $T \to S$. □

1.4. **Weil restriction.** The geometric construction of Section 4 is a space of maps from a finite flat group scheme into W, which is a derived Weil restriction. We use the following facts from [HKR].

Lemma 1.11. *Let $g\colon S' \to S$ be finite locally free and $Z' \to Z$ a quasi-smooth morphism of derived stacks over S'. Then $\mathrm{Res}_{S'/S} Z' \to \mathrm{Res}_{S'/S} Z$ is quasi-smooth, with cotangent complex*

$$\mathbb{L}_{\mathrm{Res}\, Z' / \mathrm{Res}\, Z} \simeq \left(g'_* \epsilon^* \mathbb{L}^\vee_{Z'/Z}\right)^\vee,$$

where g' is the base change of g to $\mathrm{Res}_{S'/S} Z'$ and $\epsilon\colon S' \times_S \mathrm{Res}_{S'/S} Z' \to Z'$ is the evaluation. If $Z' \to Z$ is a closed immersion, so is $\mathrm{Res}_{S'/S} Z' \to \mathrm{Res}_{S'/S} Z$.

Proof. The formula is [HKR, Proposition 2.16 and Remark A.4] and quasi-smoothness is [HKR, Corollary 2.17]: pushforward along a finite locally free map preserves perfect complexes of Tor-amplitude in $[-1, 0]$ after dualizing twice. A quasi-smooth closed immersion is of finite presentation, so its Weil restriction is a closed immersion [HKR, Theorem 5.3(ii)]. □

In particular, if $S' \to S$ has rank p and $W \to S$ is quasi-smooth of virtual dimension 0, then $\mathrm{Res}_{S'/S}(W \times_S S') \to S$ is quasi-smooth of virtual dimension 0, since the cotangent complex has Euler characteristic $p \cdot 0$.

2. A uniform exponent

The main result of this section is that, for $s > 0$ and a finitely generated field $K/\mathbb{F}_p$, the image of $\pi_{s,0} M(K) \to \pi_{s,0} \Lambda_{(p)}(K)$, $M = \mathrm{MGL}_{(p)}$, is killed by $s!$ (Proposition 2.8).

We need a bound that does not depend on the degree d of the finite syntomic algebras that represent elements of $\pi_{s,0} M(K)$. The reason is the way the bound is used in Proposition 5.2. There we show that $I_s(K)$ is p-divisible, that is $I_s(K) = p I_s(K)$. Divisibility alone does not force a group to vanish (think of $\mathbb{Q}_p/\mathbb{Z}_p$), but together with $p^e I_s(K) = 0$ it gives $I_s(K) = p^e I_s(K) = 0$. By the finite syntomic model (Theorem 2.7 below), $I_s(K)$ is the union over d of the images of classes represented by families of degree d. A bound that grows with d gives no exponent for this union. We therefore need one number e with $p^e I_s(K) = 0$, whatever the degree of the representing families. We prove $s!\, I_s(K) = 0$, hence $p^e I_s(K) = 0$ with $e = v_p(s!)$.

The proof has two parts.

(i) *Riemann–Roch without denominators.* Let $f\colon W \to Y$ be a finite syntomic family of degree d that is embedded in $\mathbb{P}^n_Y$. Then the class $f_* 1 - d$ is killed by $(n-1)!$ (Proposition 2.2). The bound depends on the dimension n of the ambient projective space and not on d. The identity has to hold integrally. A Riemann–Roch formula with rational coefficients would only determine $f_* 1 - d$ up to torsion, and the torsion is exactly what we want to bound.

(ii) *General position.* We want to apply part (1) with $n = s+1$. A finite syntomic algebra may need more than $s+1$ generators, so it need not embed in $\mathbb{A}^{s+1}$. However, the $(s+1)$-tuples of elements that fail to generate the algebra form a locus of codimension at least $s+1$ (Proposition 2.5). An s-dimensional sphere does not see such a locus, just as in topology a map from S^s to a manifold can be moved off a submanifold of codimension $s+1$. Hence every class in π_s comes from a family embedded in $\mathbb{P}^{s+1}$ (Corollary 2.6).

Finally, the finite syntomic model of MGL due to Bachmann, Elmanto, Hoyois, Khan, Sosnilo and Yakerson says that every class in $\pi_{s,0}\mathrm{MGL}(K)$ comes from such families, and this transports the bound from one family to all of $\pi_{s,0}$.

2.1. **An integral Chern class identity.** We bound $f_*1 - d$ by computing one Chern class in two ways. The identity we need is the analogue, for an MGL-algebra E with additive formal group law, of Riemann–Roch without denominators [Fu, Example 15.3.1]. For a closed immersion $i\colon Z \to Y$ of codimension n of smooth varieties, this classical result says that $c_n(i_*\mathcal{O}_Z) = (-1)^{n-1}(n-1)!\,[Z]$ in the Chow ring. For instance, if $n = 1$ then $[i_*\mathcal{O}_Z] = [\mathcal{O}_Y] - [\mathcal{O}_Y(-Z)]$, and c_1 of this class is $[Z]$. Here and below, the Chern classes of a coherent sheaf on a smooth quasi-projective scheme are those of its class in K_0, which is generated by vector bundles.

Lemma 2.1. *Let E be an additively oriented* MGL*-algebra over $\mathbb{F}_p$, and let $i\colon Z \hookrightarrow Y$ be a closed immersion of codimension $n \geq 1$ between smooth quasi-projective schemes. Then*

$$c_n([i_*\mathcal{O}_Z]) = (-1)^{n-1}(n-1)!\,i_*1 \quad \text{in } E^{2n,n}(Y).$$

Proof. We first treat the case where Z is the zero scheme of a regular section. There the Koszul complex reduces the claim to an identity of symmetric polynomials. The general case then follows by deformation to the normal bundle.

Suppose that Z is the zero scheme of a regular section of a vector bundle V of rank n. The Koszul resolution of $\mathcal{O}_Z$ gives $[i_*\mathcal{O}_Z] = \sum_r (-1)^r [\Lambda^r V^\vee]$ in $K_0(Y)$. Let $u_1, \dots, u_n$ be Chern roots of V. Since the formal group law of E is additive, the Chern roots of $\Lambda^r V^\vee$ are the sums $-\sum_{i\in I} u_i$ with $|I| = r$. Therefore

$$c([i_*\mathcal{O}_Z]) = \prod_{I\subseteq\{1,\dots,n\}} \Big(1 - \sum_{i\in I} u_i\Big)^{(-1)^{|I|}}.$$

To read off the terms of degree at most n, we take the logarithm of this power series. Since $\log(1-x) = -\sum_{m\geq 1} x^m/m$, the degree-$m$ part of the logarithm is $-\frac{1}{m}\sum_I (-1)^{|I|}(\sum_{i\in I} u_i)^m$. By inclusion–exclusion, this alternating sum is zero for $m < n$ and equals $(-1)^n n!\,u_1 \cdots u_n$ for $m = n$. Hence the product has no terms of degrees $1, \dots, n-1$, and its degree-n term is $(-1)^{n-1}(n-1)!\,u_1 \cdots u_n$. This computation takes place over $\mathbb{Q}$. What is transported to E is an identity of integral polynomials: by the splitting principle, the total Chern class of $\sum_r (-1)^r \Lambda^r V^\vee$ in an additive theory is a universal polynomial in the $c_i(V)$, and its coefficients are integral symmetric polynomials in the u_i. So the identity holds over $\mathbb{Z}$ and hence in E. Since the zero scheme of a regular section represents the top Chern class, we have $c_n(V) = i_*1$, and this proves the lemma in this case.

For a general closed immersion we deform $Z \subset Y$ to the zero section of its normal bundle. Let $\mathcal{D} = \mathrm{Bl}_{Z\times 0}(Y \times \mathbb{A}^1)$. Its fibre over 1 is Y, and its fibre over 0 is the union of $\mathrm{Bl}_Z Y$ and the exceptional divisor $P = \mathbb{P}(N_{Z/Y} \oplus \mathcal{O})$. The proper transform $j\colon Z \times \mathbb{A}^1 \hookrightarrow \mathcal{D}$ restricts to i over 1. It is disjoint from $\mathrm{Bl}_Z Y$, and it meets P transversally in the zero section of $N_{Z/Y} \subset P$, which is the zero scheme of a regular section of a vector bundle of rank n on P. Consider the difference

$$a = c_n([j_*\mathcal{O}_{Z\times\mathbb{A}^1}]) - (-1)^{n-1}(n-1)!\,j_*1$$

in $E^{2n,n}(\mathcal{D})$. By transversality, the restriction of a to P is the corresponding difference for the zero section, which vanishes by the first case. By disjointness, a restricts to zero on $\mathrm{Bl}_Z Y$.

The split blowup formula [N, Proposition 2.5 and Corollary 2.6] shows two things. First, a class that vanishes on the exceptional divisor is pulled back from $E^{2n,n}(Y \times \mathbb{A}^1) = E^{2n,n}(Y)$. Second, pullback along $\mathrm{Bl}_Z Y \to Y$ is injective. Hence a is the pullback of a class $b \in E^{2n,n}(Y)$ whose pullback to $\mathrm{Bl}_Z Y$ vanishes, and so $b = 0$. Restricting $a = 0$ to the fibre over 1 proves the lemma. □

We now apply Lemma 2.1 to a finite family embedded in a projective bundle, and compute the same Chern class a second time by the projective bundle formula. Comparing the two answers gives the bound.

Proposition 2.2. *Let E be an additively oriented* MGL*-algebra, Y a smooth quasi-projective $\mathbb{F}_p$-scheme, and $f\colon W \to Y$ finite syntomic of degree d, with W smooth over $\mathbb{F}_p$. If f factors through a closed immersion $W \hookrightarrow \mathbb{P}^n_Y$, $n \geq 1$, then*

$$(n-1)!\,(f_*1 - d) = 0 \quad \text{in } E^{0,0}(Y).$$

For $n = 1$ the proposition says that $f_*1 = d$ for every such family embedded in $\mathbb{P}^1_Y$. In general it says that the difference $f_*1 - d$ is torsion of order dividing $(n-1)!$, a bound that depends on n and not on d.

Proof. Let $\pi\colon \mathbb{P}^n_Y \to Y$ be the projection and $i\colon W \to \mathbb{P}^n_Y$ the closed immersion. The idea is to compute $\pi_* c_n([i_*\mathcal{O}_W])$ in two ways. The first computation produces $\pm(n-1)!\,f_*1$, and the second produces $\pm(n-1)!\,d$.

Since W is finite over Y, the immersion i has codimension n. Hence Lemma 2.1 gives $c_n([i_*\mathcal{O}_W]) = (-1)^{n-1}(n-1)!\,i_*1$. Applying π_*, we obtain $(-1)^{n-1}(n-1)!\,f_*1$.

For the second computation, let $h = c_1(\mathcal{O}(1))$. By the projective bundle formula for $K_0(\mathbb{P}^n_Y)$ we may write $[i_*\mathcal{O}_W] = \sum_{j=0}^n \pi^* e_j [\mathcal{O}(-j)]$ with $e_j \in K_0(Y)$. We claim that $\mathrm{rk}(e_j) = (-1)^j d\binom{n}{j}$. Ranks may be computed on fibres. Since W is flat over Y, $[i_*\mathcal{O}_W]$ restricts on fibres to $[\mathcal{O}_{W_y}]$. Over a geometric point y of Y, the fibre W_y has length d, so $[\mathcal{O}_{W_y}]$ is d times the class of a rational point. The Koszul resolution of a point cut out by n hyperplanes gives $[\mathcal{O}_{\mathrm{pt}}] = \sum_j (-1)^j \binom{n}{j} [\mathcal{O}(-j)]$, and the claim follows.

Now $c_n([i_*\mathcal{O}_W])$ is a polynomial in h with coefficients pulled back from Y, of degree at most n, because by the projective bundle formula h^{n+1} is a combination of lower powers of h with coefficients pulled back from Y; as $\mathcal{O}^{n+1}$ is trivial and the formal group law is additive, in fact $h^{n+1} = 0$. By Lemma 1.9, π_* sends it to its coefficient of h^n. This coefficient lies in $E^{0,0}(Y)$, so no Chern classes of the e_j enter it, and it depends only on the ranks $\mathrm{rk}(e_j)$. It is therefore the coefficient of t^n in

$$\prod_{j=0}^{n} (1 - jt)^{(-1)^j d\binom{n}{j}}.$$

This is the dth power of the product in the proof of Lemma 2.1 with all $u_i = t$, because exactly $\binom{n}{j}$ subsets I have $|I| = j$. By that proof, the product has no terms of degrees $1, \ldots, n-1$, so the coefficient of t^n in its dth power is d times its own coefficient of t^n, namely $(-1)^{n-1}(n-1)!\,d$. Comparing the two computations gives $(-1)^{n-1}(n-1)!\,(f_*1 - d) = 0$. □

2.2. **Generating an algebra.**

The problem. Proposition 2.2 needs a family embedded in $\mathbb{P}^n_Y$ with $n = s+1$, and the families that represent classes in π_s have arbitrary degree d. A finite syntomic algebra may need many generators. For example $k[x_1, \ldots, x_m]/(x_1^2, \ldots, x_m^2)$ has degree 2^m and a cotangent space of dimension m, so it needs m generators. Hence it cannot always be embedded in $\mathbb{A}^{s+1}$. Instead we choose an $(s+1)$-tuple of elements of the algebra and use it as an embedding where it generates. The work is to show that the non-generating tuples form a locus of codimension at least $s+1$.

For $d, m \geq 1$ let $H_{d,m} = \mathrm{Hilb}^{\mathrm{lci}}_d(\mathbb{A}^m)$ be the scheme parametrizing closed subschemes of $\mathbb{A}^m$ that are finite locally free of degree d and local complete intersections. It is smooth

of pure dimension md (Lemma 2.3 below, with $r = 0$). A point of $H_{d,m}$ is the same as a finite syntomic algebra B of degree d together with an m-tuple of elements that generates B, namely the images of the coordinates $x_1, \dots, x_m$. Thus points of Hilbert schemes are exactly algebras with a generating tuple.

The idea. Generation can be tested on quotients of length two (Lemma 2.4). This is the algebraic form of the familiar criterion for a map to be a closed embedding: it must separate points and tangent vectors. For a single length-two quotient Q of B, a tuple of n elements fails to generate Q if and only if each of its n entries maps to a scalar in Q. These are n linear conditions, so they cut out a subspace of codimension n. It remains to see that the length-two quotients do not form too large a family. We show that the space of pairs (an algebra B of degree d, a length-two quotient of B) has the same dimension as the space of algebras B alone (Lemma 2.3). Thus a union of codimension-n subspaces over this family still has codimension n. A reduced algebra has finitely many length-two quotients, one for each pair of points, and a non-reduced algebra can have a positive-dimensional family of them (see the examples after Lemma 2.4).

We first describe the flags of subschemes needed to count dimensions.

Lemma 2.3. *Let $0 \le r \le d$ and $m \ge 1$, and let $H_{r\subset d,m}$ be the Hilbert scheme of flags $C \subset B \subset \mathbb{A}^m$ with C, B finite locally free and lci of degrees r, d; for $r = 0$ we take $C = \varnothing$, so that $H_{0,m} = \operatorname{Spec} \mathbb{F}_p$ and $H_{0\subset d,m} = H_{d,m}$. The map $H_{r\subset d,m} \to H_{r,m}$ forgetting B is smooth of relative dimension $m(d-r)$. In particular, $H_{r\subset d,m}$ is smooth of pure dimension md.*

Proof. The idea is to lift B together with C by lifting the equations of B inside the ideal of the lifted C. We verify the infinitesimal lifting criterion for smoothness and then compute the tangent space.

Let $A' \to A$ be a small extension of Artin local rings, let $C \subset B$ be a flag over A, and let $C' \subset \mathbb{A}^m_{A'}$ be a lift of C. We must find a lift B' of B with $C' \subset B'$. Let $I_B \subset I_C$ be the ideals of B and C in $A[x] = A[x_1, \dots, x_m]$.

Since B is lci of codimension m and semilocal, the conormal module I_B/I_B^2 is free of rank m. Choose $f_1, \dots, f_m \in I_B$ lifting a basis. By the determinant trick there is $g \in A[x]$ with $g \equiv 1$ modulo I_B such that the f_i generate I_B in $A[x]_g$. Since C' is flat over A', the ideal of C' in $A'[x]$ surjects onto I_C, and $I_B \subset I_C$. So we can lift the f_i to elements f_i' of the ideal of C'. Lift g to some $g' \in A'[x]$. It is invertible on C', because g is invertible on C. Put

$$B' = \operatorname{Spec} A'[x]_{g'}/(f_1', \dots, f_m').$$

We check that B' is the required lift.

- It is flat over A', because the f_i' restrict to a regular sequence on the special fibre.
- It is finite over A' by Nakayama's lemma, since the maximal ideal of A' is nilpotent and $B' \otimes_{A'} A = B$ is finite. For the same reason it is generated by the images of the x_i, so B' is a closed subscheme of $\mathbb{A}^m_{A'}$, and not only of the open subscheme where $g' \neq 0$.
- It contains C', because the f_i' lie in the ideal of C' and g' is invertible on C'.
- It is a relative complete intersection of degree d, since $B' \otimes_{A'} A = B$.

Thus the forgetful map is smooth. Its relative tangent space at a flag $C \subset B$ over an algebraically closed field consists of the deformations of B that contain C, that is, of $\operatorname{Hom}_B(I_B/I_B^2, I_C/I_B)$. Since I_B/I_B^2 is free of rank m and I_C/I_B has length $d-r$, this space has dimension $m(d-r)$. For $r = 0$ this shows that $H_{d,m}$ is smooth of pure dimension md, and then for $r \ge 1$ adding $\dim H_{r,m} = mr$ gives the dimension of $H_{r\subset d,m}$. □

Lemma 2.4. *Let B be a finite algebra over an algebraically closed field. A tuple generates B if and only if its image generates every quotient algebra of B of dimension two.*

Proof. Let k be the field. If a tuple generates B, its image generates every quotient of B, so only the converse needs proof. A two-dimensional k-algebra is $k \times k$ or $k[\epsilon]/(\epsilon^2)$, and a tuple generates it if and only if some entry is not a scalar. Hence generation of all quotients $B \to k \times k$ says that the tuple separates the points of $\operatorname{Spec} B$, and generation of all quotients $B \to k[\epsilon]/(\epsilon^2)$ says that the tuple spans every cotangent space $\mathfrak{m}/\mathfrak{m}^2$ of B.

Suppose now that the image of the tuple generates every two-dimensional quotient, and let $B' \subseteq B$ be the subalgebra generated by the tuple. Since the tuple separates points, the finite map $\operatorname{Spec} B \to \operatorname{Spec} B'$ is bijective. Hence B' and B have the same idempotents, and we may assume that B is local with maximal ideal $\mathfrak{m}$. Then B' contains elements of $\mathfrak{m}$ spanning $\mathfrak{m}/\mathfrak{m}^2$, say $y_1, \dots, y_t$, so that $\mathfrak{m} = \sum_i k y_i + \mathfrak{m}^2$. Substituting this expression into itself gives $\mathfrak{m} = \sum_i k y_i + \mathfrak{m} \cdot \sum_i k y_i + \mathfrak{m}^3$ and, iterating, $\mathfrak{m} \subseteq k[y_1, \dots, y_t] + \mathfrak{m}^N$ for every N. Since $\mathfrak{m}$ is nilpotent, $\mathfrak{m}^N = 0$ for large N, so the y_i generate $B = k \oplus \mathfrak{m}$ as a k-algebra (Nakayama's lemma). □

Here are three examples over k. First, in $B = k[x]/(x^3)$ the tuple (x) generates B and the tuple (x^2) does not. The test detects this through the unique two-dimensional quotient $B \to k[x]/(x^2)$, under which x^2 maps to 0, a scalar, while x does not.

Second, let $B = k \times k$. A tuple $(b_1, \dots, b_n)$ with $b_i = (\beta_i, \gamma_i)$ generates B if and only if $\beta_i \neq \gamma_i$ for some i. The non-generating tuples form the linear subspace $\beta = \gamma$ of codimension n in $B^{\oplus n} = k^{2n}$.

Third, let $B = k[x, y]/(x^2, y^2)$, a complete intersection of degree four with maximal ideal $\mathfrak{m} = (x, y)$ and $\mathfrak{m}/\mathfrak{m}^2$ of dimension two. Its two-dimensional quotients are the algebras $B/(\ell + \mathfrak{m}^2) \cong k[\epsilon]/(\epsilon^2)$, where ℓ runs through the lines in $\mathfrak{m}/\mathfrak{m}^2$. They form a family parametrized by $\mathbb{P}^1$. An element of B maps to a scalar in $B/(\ell + \mathfrak{m}^2)$ exactly when its linear part lies in ℓ. Hence an n-tuple generates B if and only if the linear parts of its entries span $\mathfrak{m}/\mathfrak{m}^2$. The non-generating n-tuples form a locus of codimension $n-1$ in $B^{\oplus n}$, one less than for $k \times k$, because the quotients vary in a family of dimension one. This does not contradict Proposition 2.5 below: such algebras themselves lie in a stratum of positive codimension in U_d, and the count there is over the total space.

The stack of finite syntomic algebras. To count codimensions and to apply purity we need smooth schemes, so we present the stack FSyn_d of finite syntomic algebras of degree d by smooth schemes and describe $H_{d,n}$ over it.

- Let $U_d \to \mathrm{FSyn}_d$ be the scheme of finite syntomic algebras of degree d with a basis whose first element is 1. It is a torsor under $G_d = \{g \in \mathrm{GL}_d : ge_1 = e_1\}$. Since G_d is an extension of GL_{d-1} by a vector group, G_d-torsors are Zariski locally trivial.
- For $d \geq 2$, writing the basis as $1, x_1, \dots, x_{d-1}$ identifies U_d with the open subscheme of $H_{d,d-1}$ where $1, x_1, \dots, x_{d-1}$ is a basis. Hence U_d is smooth, since $H_{d,d-1}$ is.
- The universal algebra $W_d = \operatorname{Spec} B_d \to U_d$ is the corresponding open part of $H_{1 \subset d,d-1}$, since a point of W_d over U_d is a degree-one closed subscheme (a section) of the fibre. Hence W_d is smooth by Lemma 2.3.
- The Čech nerve $U_d \times G_d^\bullet$ presents FSyn_d by smooth schemes.
- The n-tuples of elements of the universal algebra form the vector bundle $\mathrm{Tot}(B_d^{\oplus n})$ over FSyn_d, and $H_{d,n}$ is its open substack of generating tuples.

Proposition 2.5. *The complement of $H_{d,n}$ in $\mathrm{Tot}(B_d^{\oplus n})$ has codimension at least n, at every level of the Čech nerve.*

Proof. The idea is to cover the non-generating locus by a vector bundle over the space of length-two quotients, using Lemma 2.4, and to compare dimensions.

For $d \leq 1$ every tuple generates, so the complement is empty. Let $d \geq 2$. The levels $U_d \times G_d^k$ of the Čech nerve are smooth over U_d, and codimension is preserved under smooth pullback, so it suffices to work over U_d. There, $\mathrm{Tot}(B_d^{\oplus n})$ has dimension $\dim U_d + nd$.

Let $Z_d \to U_d$ be the relative Hilbert scheme of quotients of B_d of length two. Every algebra of rank two is syntomic, since it has a basis $1, z$ and is cut out by one monic quadratic relation. Hence Z_d is the open part of $H_{2\subset d,d-1}$ lying over U_d, and by Lemma 2.3 it is smooth of dimension $d(d-1) = \dim U_d$. Let Q be the universal quotient algebra on Z_d, a locally free sheaf of rank two containing $\mathcal{O}$. A tuple in $B_d^{\oplus n}$ fails to generate Q exactly when its image in $(Q/\mathcal{O})^{\oplus n}$ is zero. The kernel of the surjection $B_d^{\oplus n} \to (Q/\mathcal{O})^{\oplus n}$ is therefore a vector bundle of rank $n(d-1)$ on Z_d. Its total space parametrizes pairs of a length-two quotient and a tuple that fails to generate it. By Lemma 2.4, the image of this total space in $\mathrm{Tot}(B_d^{\oplus n})$ is exactly the non-generating locus. Hence the non-generating locus has dimension at most $\dim U_d + n(d-1)$, that is, codimension at least n. □

From codimension to connectivity. Proposition 2.5 becomes a statement in homotopy theory through homotopy purity. If F is a smooth closed subscheme of codimension c in a smooth scheme X, then $X/(X \smallsetminus F)$ is the Thom space of the normal bundle of F, and its S^1-suspension spectrum is c-connective. So removing a locus of codimension at least n does not lose classes in degrees below n: the map on π_i is an isomorphism for $i < n-1$ and surjective for $i = n-1$. Applied with $n = s+1$, the following corollary says that, for the purpose of producing every class in π_s, the stack FSyn_d of all finite syntomic algebras of degree d may be replaced by the space $H_{d,n}$ of algebras with a generating n-tuple.

Corollary 2.6. *The cofibre $C_{d,n}$ of*

$$\Sigma^\infty_{S^1} H_{d,n,+} \longrightarrow \Sigma^\infty_{S^1} \mathrm{FSyn}_{d,+}$$

belongs to $\mathrm{SH}^{S^1}(\mathbb{F}_p)_{\geq n}$.

Proof. The vector bundle $\mathrm{Tot}(B_d^{\oplus n}) \to \mathrm{FSyn}_d$ is an $\mathbb{A}^1$-equivalence, so $C_{d,n}$ is the cofibre of the inclusion of $H_{d,n}$ into $\mathrm{Tot}(B_d^{\oplus n})$. We compute this cofibre levelwise in the Čech nerve, where all objects are smooth schemes, and then take the geometric realization.

At each level, let F be the closed non-generating locus. Since $\mathbb{F}_p$ is perfect, the reduced scheme F admits a stratification by smooth locally closed subschemes, and by Proposition 2.5 every stratum has codimension at least n. Removing the strata one at a time, we write the cofibre as an iterated extension of cofibres $X/(X \smallsetminus F_i)$, where F_i is a smooth closed stratum of a smooth scheme X. By homotopy purity, each $X/(X \smallsetminus F_i)$ is the Thom space of the normal bundle of F_i, which has rank at least n. The S^1-suspension spectrum of the Thom space of a vector bundle of rank c on a smooth scheme X is c-connective. Indeed, the bundle is Zariski locally trivial, and the Thom space of the trivial bundle is $S^c \wedge \mathbb{G}_m^{\wedge c} \wedge X_+$. Connectivity is Zariski local: the Thom space is the colimit of the Čech diagram of the Thom spaces over a trivializing cover, and colimits preserve connectivity. Extensions and geometric realizations preserve n-connectivity, so $C_{d,n}$ is n-connective. □

2.3. **The factorial bound on homotopy groups.**

The finite syntomic model of MGL. It remains to see that the classes $f_*1 - d$ of finite syntomic families account for all of $\pi_{s,0}\mathrm{MGL}(K)$. A finite syntomic morphism $f\colon W \to X$ has a fundamental class $f_*1 \in \mathrm{MGL}^{0,0}(X)$. These classes assemble into a map of presheaves $\mathrm{fc}\colon \mathrm{FSyn} \to \Omega^\infty_{\mathbb{P}^1}\mathrm{MGL}$ [EHKSY, Remark 3.3.15]. Let $\mathrm{FSyn}_\infty = \mathrm{colim}_d \mathrm{FSyn}_d$, where the transition maps add a copy of the base to an algebra, $B \mapsto B \times \mathcal{O}$. A family of degree d and its image in degree $d+1$ have fundamental classes differing by 1, so the class $\mathrm{fc}(W) - d$ is compatible with the transition maps. Let $L_{\mathbb{A}^1}$ denote naive simplicial $\mathbb{A}^1$-localization, that is, $(L_{\mathbb{A}^1}F)(X) = \mathrm{colim}_{m\in\Delta^{\mathrm{op}}} F(X \times \mathbb{A}^m)$. The following result says that, over a field of positive characteristic, these classes give all of $\Omega^\infty_{\mathbb{P}^1}\mathrm{MGL}$.

Theorem 2.7 ([BEHKSWY, Theorem 1.1, Theorem 1.2 and Proposition 5.6])**.** *Let K be a field of characteristic $p > 0$.*

(i) *There is an equivalence of spaces*

$$(\Omega^\infty_{\mathbb{P}^1}\mathrm{MGL})(K) \simeq \mathbb{Z} \times (L_{\mathbb{A}^1}\mathrm{FSyn}_\infty)(K), \qquad (a, W) \longmapsto a + \mathrm{fc}(W) - d, \tag{2.1}$$

where d is the degree of W.

(ii) *There is an equivalence of spaces*

$$(\Omega^\infty_{\mathbb{P}^1}\mathrm{MGL})(K) \simeq \mathbb{Z} \times \big(L_{\mathbb{A}^1}\,\mathrm{Hilb}^{\mathrm{lci}}_\infty(\mathbb{A}^\infty)\big)(K), \qquad (a, W) \longmapsto a + \mathrm{fc}(W) - d,$$

where $\mathrm{Hilb}^{\mathrm{lci}}_\infty(\mathbb{A}^\infty)$ is the colimit of the Hilbert schemes $H_{d,m}$ over m and over d, along the closed immersions that add a point.

Over an arbitrary field, [BEHKSWY] proves these equivalences after applying Quillen's plus construction to the right hand sides; in positive characteristic the plus construction is not needed [BEHKSWY, Proposition 5.6], and this is why we can read off homotopy groups of MGL directly from finite syntomic families. The two parts correspond to each other because the map $\mathrm{colim}_m H_{d,m} \to \mathrm{FSyn}_d$ that forgets the embedding is a motivic equivalence [BEHKSWY], so both models of Theorem 2.7 agree; below we use whichever is more convenient. Part (2) is used in Proposition 5.2.

We use part (1) as follows. Let $s > 0$. The factor $\mathbb{Z}$ is discrete, so it contributes nothing to π_s, and π_s commutes with the filtered colimit over d. Consequently every element of $\pi_{s,0}\mathrm{MGL}(K)$ comes from $\pi_s((L_{\mathbb{A}^1}\mathrm{FSyn}_d)(K))$ for some d and some base point. Its image in $\pi_{s,0}\Lambda_{(p)}(K)$ is the image under the map (L_{mot} as in the Conventions)

$$\delta_d \colon L_{\mathrm{mot}}\mathrm{FSyn}_d \longrightarrow \Omega^\infty_{\mathbb{P}^1}\Lambda_{(p)}, \qquad W \longmapsto f_*1 - d.$$

Here f_*1 denotes the fundamental class with coefficients in $\Lambda_{(p)}$, so $\delta_d = \rho \circ \mathrm{fc} - d$ is first a map of presheaves on FSyn_d; it factors through $L_{\mathrm{mot}}\mathrm{FSyn}_d$ since the target is motivically local. A class of $\pi_s((L_{\mathbb{A}^1}\mathrm{FSyn}_d)(K))$ is sent to π_s of $(L_{\mathrm{mot}}\mathrm{FSyn}_d)(K)$ by the canonical map $(L_{\mathbb{A}^1}\mathrm{FSyn}_d)(K) \to (L_{\mathrm{mot}}\mathrm{FSyn}_d)(K)$, and then by δ_d. Hence it suffices to bound the map induced by δ_d on π_s, by a number that does not depend on d.

The bound. The proof of Proposition 2.8 combines Proposition 2.2 on $H_{d,n}$, $n = s + 1$, with Corollary 2.6 for its complement.

Proposition 2.8. *For $s > 0$ and every finitely generated field $K/\mathbb{F}_p$, the image*

$$I_s(K) = \mathrm{im}\big(\pi_{s,0}\mathrm{MGL}_{(p)}(K) \longrightarrow \pi_{s,0}\Lambda_{(p)}(K)\big)$$

is annihilated by $s!$.

Proof. Set $n = s + 1$. Over $H_{d,n}$, the generating n-tuple embeds the universal family in $\mathbb{A}^n \subset \mathbb{P}^n_{H_{d,n}}$, and the total space of this family is $H_{1\subset d,n}$, which is smooth by Lemma 2.3. Since $\Lambda_{(p)}$ has additive formal group law, Proposition 2.2 applies with $(n-1)! = s!$ and shows that $s!\,\delta_d$ restricts to zero on $H_{d,n}$.

To pass to homotopy groups we use spectra. Let $\Omega^\infty_{\mathbb{G}_m}\Lambda_{(p)}$ denote the underlying motivic S^1-spectrum of $\Lambda_{(p)}$, so that δ_d is adjoint to a map

$$u_d \colon \Sigma^\infty_{S^1}\mathrm{FSyn}_{d,+} \longrightarrow \Omega^\infty_{\mathbb{G}_m}\Lambda_{(p)}.$$

The map $s!\,u_d$ restricts to zero on $\Sigma^\infty_{S^1}H_{d,n,+}$, and hence it factors through the cofibre $C_{d,n}$, and, the target being motivically local, through $L_{\mathrm{mot}}C_{d,n}$. By Corollary 2.6, $C_{d,n}$ is $(s+1)$-connective, and since K is an essentially smooth field over $\mathbb{F}_p$, $\pi_s(L_{\mathrm{mot}}C_{d,n})(K) = \underline{\pi}_s(C_{d,n})(K) = 0$. Hence $s!\,u_d$ is zero on π_s at K. The map induced by δ_d on π_s factors through the stabilization map into $\pi_s(\Sigma^\infty_{S^1}\mathrm{FSyn}_{d,+}(K))$ followed by u_d, so $s!\,\delta_d$ is zero on π_s at every base point.

By Theorem 2.7(1), every element of $\pi_{s,0}\mathrm{MGL}(K)$ maps into $\pi_{s,0}\Lambda_{(p)}(K)$ through some δ_d. Hence $s!$ kills the image of $\pi_{s,0}\mathrm{MGL}(K)$. Since $\pi_{s,0}\mathrm{MGL}_{(p)}(K) = \pi_{s,0}\mathrm{MGL}(K) \otimes \mathbb{Z}_{(p)}$, the same holds for $I_s(K)$. □

3. Prime-to-p covers and normalized fundamental classes

The proof of the necklace formula in Section 4 needs the fundamental classes of spaces that are stacks rather than schemes. The spaces built in Section 4 are quotient stacks with finite stabilizers of order prime to p. They are quotients by $\mathbb{G}_m$ (to make them proper) and by a group of order p (which produces the factor p in $d^p - d$); see Section 4.

Theorem 1.6, on the other hand, provides fundamental classes only for quasi-smooth morphisms of derived schemes. We do not develop pushforward along stacks. Instead, this section explains how to attach a class $\langle S \to X \rangle$ to a tame stack S that is proper and quasi-smooth over a smooth affine scheme X, using only fundamental classes of derived schemes. Throughout this section fundamental classes have coefficients in $\Lambda_{(p)}$.

The idea. Classically, one computes with an orbifold by passing to a cover. If a finite group G acts on a smooth projective variety V, the orbifold Euler characteristic of $[V/G]$ is $\chi(V)/|G|$, since $V \to [V/G]$ is a finite cover of degree $|G|$. We imitate this. Suppose $a\colon Z \to S$ is finite flat of rank r, with Z a derived scheme. If S had a class $\langle S \to X \rangle$ and pushforward along $S \to X$ were functorial, then $[Z \to X]$ would be the pushforward of a_*1. So if $a_*1 = r$, we would have $[Z \to X] = r\langle S \to X \rangle$. We therefore define $\langle S \to X \rangle$ as $r^{-1}[Z \to X]$. For this to make sense three things are needed.

(i) The rank r must be prime to p, so that r is invertible in $\Lambda^{0,0}_{(p)}(X)$, which is a $\mathbb{Z}_{(p)}$-module.
(ii) The trace a_*1 must equal r, after every base change to a derived scheme.
(iii) The class $r^{-1}[Z \to X]$ must not depend on the cover.

Condition (iii) is Lemma 3.3 below, and its proof is a short argument with the fibre product of two covers. The existence of covers satisfying (i) and (ii) is Proposition 3.5.

Condition (i) explains why only tame stacks occur. If $Z \to S$ is representable and finite flat of rank r, with Z an algebraic space, and x is a geometric point of S with stabilizer G_x, then the classical fibre of Z over x is a finite scheme of length r on which G_x acts freely, so the order of G_x divides r. Hence a cover of prime to p rank can only exist if all stabilizers have order prime to p. Conversely, a finite group scheme of order prime to p is étale and linearly reductive, so such stacks are tame. The stacks $\mathfrak{B}$ and Y of Section 4 have this property because the weights $1, \dots, p-1$ are prime to p.

Why the trace has to be checked. For a finite flat morphism $a\colon Z \to S$ of rank r between derived schemes, the trace a_*1 is a class in $\Lambda^{0,0}_{(p)}(S)$, and there is no general reason for it to be the integer r. Indeed, for the finite syntomic morphism $f\colon W \to X$ of degree d in Theorem 4.1, the trace f_*1 is exactly the unknown class of that theorem. If it were always equal to d, the congruence $(f_*1)^p \equiv f_*1 \pmod p$ in $\Lambda_{(p)}$ would be trivial.

The traces we can compute are those of Corollary 1.10. If Z is the derived zero locus, in a projective bundle over S, of a section of a sum of line bundles $\mathcal{O}(e_i) \otimes \pi^* A_i$, and Z is finite over S, then its trace is $e_1 \cdots e_r$, after every derived base change. This rests on the additive formal group law of $\Lambda_{(p)}$. All covers in this paper are cut out by monic equations, which can be put in this form. The basic case is the extraction of a root.

Lemma 3.1. *Let S be a derived scheme, L a line bundle on S, $w \geq 1$ an integer and $s \in \Gamma(S, L^w)$, and let $h\colon Z = \{z \in \mathrm{Tot}(L) : z^w = s\} \to S$. Then h is finite locally free of rank w and quasi-smooth of virtual relative dimension zero, with*

$$h_*\mathcal{O}_Z \simeq \bigoplus_{j=0}^{w-1} L^{-j}$$

as $\mathcal{O}_S$-modules. Moreover $[Z \to S] = w$ in $\Lambda^{0,0}_{(p)}(S)$, compatibly with every derived base change.

Proof. On an affine open trivializing L, write the equation as $t^w - s = 0$ over an animated ring R. Multiplication by this monic polynomial is injective on every $\pi_j(R)[t]$. The two-term Koszul complex for its derived zero locus therefore shows that the quotient has homotopy freely generated over $\pi_* R$ by $1, t, \dots, t^{w-1}$. These monomials define a global map $\bigoplus_{j=0}^{w-1} L^{-j} \to h_* \mathcal{O}_Z$, which is an equivalence by the local calculation. Thus h is finite locally free of rank w; since $w > 0$, it is also surjective. Its single equation in the smooth line bundle $\mathrm{Tot}(L) \to S$ makes it smoothable and quasi-smooth of virtual relative dimension zero.

To compute its trace, let x and z be the coordinates of $\mathbb{P}(\mathcal{O} \oplus L)$ corresponding to $\mathcal{O}$ and L, so that $x = 0$ is the section at infinity $\mathbb{P}(L)$. The homogeneous polynomial $z^w - sx^w$ is a section of $\mathcal{O}(w) \otimes \pi^* L^w$. Its derived zero locus avoids infinity, where $x = 0$ and $z \neq 0$, and on $x \neq 0$ it is precisely Z. Hence Corollary 1.10 gives $[Z \to S] = w$. The zero locus, its module decomposition, and this projective-bundle calculation commute with derived base change, so the same trace identity holds after every such base change. □

Over the locus where s is invertible, Z is a μ_w-torsor, and over a geometric point where $s = 0$ its fibre is $\operatorname{Spec} \kappa[z]/(z^w)$, which is nonreduced when $w > 1$. The trace computation does not distinguish the two loci.

Good covers and normalized classes.

Definition 3.2. A *good cover* of a derived stack S is a representable finite flat quasi-smooth surjection $a\colon Z \to S$, with Z a derived scheme, of constant rank r prime to p, such that for every map $S' \to S$ from a derived scheme the base change $a_{S'}\colon Z \times_S S' \to S'$ has trace $(a_{S'})_* 1 = r$ in $\Lambda^{0,0}_{(p)}(S')$.

Good covers are stable under composition, since traces are multiplicative by Corollary 1.7(iii), and under base change along maps $S' \to S$ for which $Z \times_S S'$ is again a derived scheme. The trace condition is not automatic, as explained above, so we verify it for each cover we use. In every case it holds because the cover is cut out by monic equations as in Lemma 3.1.

The next lemma is condition (iii) above. The idea is that two good covers Z and Z' of S are dominated by their fibre product $Z \times_S Z'$, which is a good cover of each of them, so both normalized classes equal the normalized class of the fibre product. The ample line bundles in the hypothesis make the maps involved smoothable (Lemma 1.8) and therefore have fundamental classes.

Lemma 3.3. *Suppose $S \to X$ is proper and quasi-smooth, with X smooth affine, and $Z \to S$, $Z' \to S$ are good covers of ranks r, r'. Assume that Z and Z' carry line bundles ample on their classical truncations. Let $S_v \subseteq S$ be the open-and-closed locus where $S \to X$ has virtual relative dimension v, and put $Z_v = Z \times_S S_v$ and $Z'_v = Z' \times_S S_v$. Then*

$$r'[Z_v \to X] = r[Z'_v \to X] \quad \text{in } \Lambda^{-2v,-v}_{(p)}(X)$$

for every v. Consequently the normalized class

$$\langle S \to X \rangle := \sum_v r^{-1}[Z_v \to X] \quad \text{in } \bigoplus_{v \in \mathbb{Z}} \Lambda^{-2v,-v}_{(p)}(X) \tag{3.1}$$

is independent of the good cover. If $S \to X$ has constant virtual relative dimension v, it is the homogeneous class $r^{-1}[Z \to X] \in \Lambda^{-2v,-v}_{(p)}(X)$.

Proof. The virtual rank of the perfect cotangent complex is locally constant. Only finitely many S_v are nonempty, since S is quasi-compact. Restricting the covers and their ample line bundles to these open-and-closed loci preserves the hypotheses and the ranks r, r'. It therefore suffices to prove the assertion when $S \to X$ has constant virtual relative dimension v.

Let $P = Z \times_S Z'$. Its projection to Z is finite, so P is a derived scheme. The pullback of the given line bundle on Z is ample on $t_0 P$, since $t_0 P \to t_0 Z$ is finite. The scheme maps

$P \to Z$, $P \to Z'$, $Z \to X$, $Z' \to X$ and $P \to X$ are proper and quasi-smooth. Their sources carry actual line bundles ample on their classical truncations, and their targets are separated over X. Thus Lemma 1.8 makes these maps smoothable, and their fundamental classes are defined and compatible with composition.

Write $q\colon P \to Z$ and $g\colon Z \to X$. The map q is the base change of the good cover $Z' \to S$ along $Z \to S$, so $q_*1 = r'$. A finite flat quasi-smooth map has virtual relative dimension zero: its geometric fibres are classical zero-dimensional local complete intersections. Thus $P \to X$ and $Z \to X$ have virtual dimension v. The composition rule gives

$$[P \to X] = g_*q_*1 = r'[Z \to X] \quad \text{in } \Lambda_{(p)}^{-2v,-v}(X).$$

Similarly $[P \to X] = r[Z' \to X]$. Dividing by rr', which is invertible in the p-local coefficient ring, proves independence on this virtual-dimension locus. Summing the finitely many homogeneous classes proves independence of (3.1). □

Thus (3.1) defines the class of a stack without a separate theory of pushforward on stacks. In Section 4 every class of a stack is computed in this way, on a good cover by a scheme, and then divided by the rank of the cover. When S is itself a derived scheme with an ample line bundle on t_0S, the identity $S \to S$ is a good cover of rank 1, so $\langle S \to X\rangle = [S \to X]$, interpreted componentwise by virtual dimension. If $Z \to S$ is a good cover and $D \to S$ is a map such that $D_Z = D \times_S Z$ is a derived scheme with an ample line bundle, then $D_Z \to D$ is a good cover of the same rank. Whenever $\langle D \to X\rangle$ is defined, it may therefore be computed on D_Z. This is how Lemma 4.7 is proved. In the applications to E, D and Y in Section 4, the virtual relative dimension over X is zero, so all these normalized classes belong to $\Lambda_{(p)}^{0,0}(X)$.

3.1. **Root covers.** The basic example of a good cover is a root cover of a weighted projective stack. Recall that for positive integers $w_0, \dots, w_m$ the weighted projective stack is the quotient

$$\mathbb{P}(w_0, \dots, w_m) = [(\mathbb{A}^{m+1} \setminus 0)/\mathbb{G}_m], \qquad \lambda \cdot (y_0, \dots, y_m) = (\lambda^{w_0}y_0, \dots, \lambda^{w_m}y_m).$$

The stabilizer of a point y is μ_g, where g is the greatest common divisor of the w_i with $y_i \neq 0$. A map from a derived scheme S' to $\mathbb{P}(w_0, \dots, w_m)$ is a line bundle $\mathcal{L}$ on S' together with sections $s_i \in \Gamma(S', \mathcal{L}^{w_i})$ that have no common zero, and $\mathcal{O}(1)$ pulls back to $\mathcal{L}$.

The simplest case is $\mathbb{P}^1 \to \mathbb{P}(1,2)$, $[z_0 : z_1] \mapsto [z_0 : z_1^2]$, induced by the $\mathbb{G}_m$-equivariant map $(z_0, z_1) \mapsto (z_0, z_1^2)$ of $\mathbb{A}^2 \setminus 0$ (weights $1,1$ on the source, $1,2$ on the target). On the atlas it is finite flat of rank 2. Over the stacky point $[0 : 1]$, with stabilizer μ_2, the atlas fibre is $\{(0, \pm 1)\}$, and μ_2 exchanges the two points, so it acts freely on the fibre of the cover. The rank 2 is divisible by $|\mu_2|$, as it must be. If $p \neq 2$, this is a good cover by the general statement below.

In general, for positive weights $w_0, \dots, w_m$ prime to p, the map

$$\mathbb{P}^m \longrightarrow \mathbb{P}(w_0, \dots, w_m), \qquad [z_i] \longmapsto [z_i^{w_i}] \tag{3.2}$$

is a good cover of rank $\prod_i w_i$. It is representable since $\mathbb{P}^m$ has trivial stabilizers, and its source is a scheme. To check the rank and the trace, consider the base change along a map from a derived scheme S', given by $\mathcal{L}$ and sections s_i of $\mathcal{L}^{w_i}$ as above. The map (3.2) is $\mathbb{G}_m$-equivariant for the identity of $\mathbb{G}_m$, so it pulls back $\mathcal{O}(1)$ to $\mathcal{O}(1)$, and a lift of $(\mathcal{L}, s_i)$ to $\mathbb{P}^m$ is exactly a tuple of sections z_i of $\mathcal{L}$ with $z_i^{w_i} = s_i$. The base change is cut out of $\mathrm{Tot}(\mathcal{L}^{\oplus(m+1)})$ by the equations $z_i^{w_i} = s_i$, where z_i is a section of $\mathcal{L}$, and the z_i have no common zero because the s_i have none. Each equation $z_i^{w_i} = s_i$ defines a cover of S' as in Lemma 3.1, finite locally free of rank w_i with trace w_i. The base change of (3.2) is the fibre product over S' of these $m+1$ covers. Its rank is $\prod_i w_i$. Its trace is $\prod_i w_i$ as well, because pushforward is functorial and each factor has trace w_i after every derived base change. Since S' was arbitrary, this proves the trace identity after every base change.

The same construction, taking roots of sections of a line bundle, produces good covers of more general tame stacks. What makes (3.2) representable is that the stabilizers act freely on the roots. In the example of $\mathbb{P}^1 \to \mathbb{P}(1,2)$ above, the stabilizer μ_2 of $[0:1]$ acts on the root z_1 of y_1 through the inclusion $\mu_2 \subseteq \mathbb{G}_m$, so the nonzero vector z_1 has trivial stabilizer. To have the same effect on a general stack, we need a line bundle on which the stabilizers act through injective characters.

Definition 3.4. A line bundle on a Deligne–Mumford stack is *faithful* if every geometric stabilizer acts on its fibre through an injective character.

For example, $\mathcal{O}(1)$ is faithful on $\mathbb{P}(w_0,\dots,w_m)$, since the stabilizer μ_g acts on its fibre through the inclusion $\mu_g \subseteq \mathbb{G}_m$. On the stack Y of Section 4 the line bundle $\mathcal{L}$ of weight -1 is faithful; this is verified in Lemma 4.5.

The point of faithfulness is the following. Let x be a geometric point of a stack S with stabilizer G_x, let $\mathcal{A}$ be a faithful line bundle, and let u be a nonzero vector in the fibre $\mathcal{A}_x$. An element $\gamma \in G_x$ sends u to $\chi(\gamma)u$, where χ is the character of G_x on $\mathcal{A}_x$. Since χ is injective, γ fixes u only if $\gamma = 1$. So G_x acts freely on the nonzero vectors of $\mathcal{A}_x$, and a space of roots of nonvanishing sections of a power of $\mathcal{A}$ has trivial stabilizers.

Covers of Deligne–Mumford stacks by schemes are constructed in general by Kresch and Vistoli [KV]. We use the explicit root construction below because its monic equations also determine the trace.

3.2. **Good covers of tame stacks.** The following proposition produces the good cover of the necklace stack Y in Lemma 4.5. Before stating it we describe the construction. Let S be a tame Deligne–Mumford stack with a faithful line bundle $\mathcal{A}$, and let e be prime to p. Choose sections $s_1,\dots,s_N$ of $\mathcal{A}^e$ with no common zero, and let Z be the space of simultaneous eth roots, that is, of tuples $(u_1,\dots,u_N)$ of vectors in $\mathcal{A}$ with $u_i^e = s_i$. Then $Z \to S$ is finite flat with trace e^N, Z has trivial stabilizers by faithfulness, and Z is a scheme; this is checked in the last step of the proof of Proposition 3.5. The work in the proof consists in arranging that the sections exist and lift from t_0S to the derived stack S. For this we need a power of $\mathcal{A}$ that is the pullback of an ample bundle on the coarse space, which is the reason for the second line bundle $\mathcal{M}$ in the hypotheses.

Proposition 3.5. *Let X be affine of finite type over a field of characteristic p. Let S be a separated tame derived Deligne–Mumford stack of finite type over X whose classical coarse space S_c is projective over X. Suppose that S carries*

(i) *a faithful line bundle $\mathcal{L}$, and*
(ii) *a line bundle $\mathcal{M}$ whose restriction to t_0S has a prime-to-p power descending to an ample line bundle on S_c.*

Suppose also that the sheaves $\pi_j\mathcal{O}_S$ are coherent and vanish for $j \gg 0$. Then S has a good cover $Z \to S$, and Z carries a line bundle ample on t_0Z.

In our application X is smooth affine over $\mathbb{F}_p$ and S is proper and quasi-smooth over X.

Proof. We follow the construction described above, in three steps: we choose a faithful line bundle $\mathcal{A}$ some power of which descends to an ample bundle, we choose and lift the sections, and we check that the space of roots is a good cover by a scheme.

Choice of the line bundle. Choose an integer c prime to p that kills all geometric stabilizers. It exists because S is tame, so its stabilizers have order prime to p, and S is a quasi-compact Deligne–Mumford stack of finite type, so the orders of its geometric stabilizers are bounded (the inertia is finite and unramified over the quasi-compact S). Then $\mathcal{L}^c|_{t_0S}$ descends to S_c, since the stabilizers act trivially on its fibres [FO, Lemma 2.3]. Suppose that $\mathcal{M}^a|_{t_0S}$ descends to an ample line bundle, with a prime to p. For $b \gg 0$ put

$$\mathcal{A} = \mathcal{L} \otimes \mathcal{M}^{ab}.$$

We note two properties of $\mathcal{A}$. The stabilizers act trivially on $\mathcal{M}^a$, because it descends to S_c. Hence $\mathcal{A}$ has the same stabilizer characters as $\mathcal{L}$, and it is faithful. Moreover, $\mathcal{A}^c|_{t_0S}$ is the tensor product of the descended bundle $\mathcal{L}^c$ with the bc-th power of the descended ample bundle $\mathcal{M}^a$. For b large it therefore descends to an ample line bundle on S_c.

Choice of the sections. Choose a sufficiently large multiple e of c with $p \nmid e$. Then the descended bundle $\mathcal{A}^e|_{t_0S}$ is globally generated, so it has finitely many sections $s_1, \dots, s_N$ with no common zero. We claim that these sections lift from t_0S to S. Global sections of $\mathcal{A}^e$ on S are computed by a descent spectral sequence whose E_2 terms are the cohomology groups of the sheaves $\pi_j\mathcal{O}_S \otimes \mathcal{A}^e$ on t_0S. Since S is tame, pushforward ϖ_* from t_0S to S_c is exact on quasi-coherent sheaves (this is what tameness provides). Write $e = ck$; since $\mathcal{A}^c|_{t_0S}$ descends to an ample bundle $\mathcal{A}_c$ on S_c, the projection formula gives $\varpi_*(\pi_j\mathcal{O}_S \otimes \mathcal{A}^e) = \varpi_*(\pi_j\mathcal{O}_S) \otimes \mathcal{A}_c^k$. So these groups are cohomology groups on the projective scheme S_c of the coherent sheaves $\varpi_*(\pi_j\mathcal{O}_S)$ twisted by $\mathcal{A}_c^k$. Only finitely many $\pi_j\mathcal{O}_S$ are nonzero, so only finitely many coherent sheaves occur, and a single Serre bound shows that all the higher cohomology groups vanish once e is large enough. The spectral sequence then degenerates and gives $\pi_0\Gamma(S, \mathcal{A}^e) = \Gamma(t_0S, \mathcal{A}^e)$, which proves the claim. This is the argument of Lemma 1.8, carried out on the coarse space.

The space of roots. Now we take all the roots at once. Let

$$Z = \{(u_1, \dots, u_N) \in \mathrm{Tot}(\mathcal{A}^{\oplus N}) : u_i^e = s_i \text{ for every } i\}. \tag{3.3}$$

Each equation $u_i^e = s_i$ is monic in u_i, so $Z \to S$ is representable, finite flat and quasi-smooth of rank e^N. It is surjective since a root cover is surjective. We claim that Z is a derived scheme. At a geometric point of Z, some s_i is nonzero, because the s_i have no common zero, so the corresponding u_i is a nonzero vector in the fibre of $\mathcal{A}$. The stabilizer of the point fixes this vector, and since it acts on that fibre through an injective character, it is trivial. Hence Z is a derived algebraic space. The classical map $t_0Z \to t_0S \to S_c$ is proper and quasi-finite, hence finite, and an algebraic space finite over a scheme is a scheme. Consequently Z is a derived scheme. The restriction of $\mathcal{A}^e$ to t_0Z is the pullback, along the finite map $t_0Z \to S_c$, of an ample line bundle on S_c, so it is ample.

It remains to check the trace. After base change along any map from a derived scheme, Z is the fibre product of N covers cut out by monic equations $u_i^e = s_i$. By Lemma 3.1 each of them has trace e, so the base change of Z has trace e^N, as in the proof for (3.2). Since e is prime to p, this proves that $Z \to S$ is a good cover. □

4. The necklace construction and the power congruence

The divisibility argument of Section 5 rests on a geometric form of Fermat's congruence $d^p \equiv d \pmod p$. Write $\rho\colon \mathrm{MGL}_{(p)} \to \Lambda_{(p)}$ for the quotient map. For a finite syntomic morphism $f\colon W \to X$ the class f_*1 plays the role of the degree d, and we show that $(f_*1)^p - f_*1$ is divisible by p in $\Lambda_{(p)}$. For Proposition 5.2 divisibility alone is not enough: we need a class with this p-multiple that lifts along ρ to cobordism. We construct such a class as the normalized fundamental class $\langle Y \to X\rangle$ of the stack Y of non-constant necklaces in W. Away from a degenerate locus, Y is the quotient of the complement of the diagonal in W^p by cyclic rotation. Its definition is given in Section 4.5.

Theorem 4.1 (Geometric necklace formula)**.** *Let X be smooth affine over $\mathbb{F}_p$, let $f\colon W \to X$ be finite syntomic, and set $u = f_*^{\mathrm{MGL}_{(p)}}1$. Let Y be the stack of non-constant necklaces in W, constructed in Section 4.5. Then Y has a good cover $Z \to Y$ of some rank r, and the class $b = r^{-1}[Z \to X]_{\mathrm{MGL}_{(p)}}$ in $\mathrm{MGL}_{(p)}^{0,0}(X)$ satisfies*

$$\rho(u^p - u) = p\rho(b). \tag{4.1}$$

Here $\rho(b)$ is the normalized class $\langle Y \to X\rangle$ in $\Lambda^{0,0}_{(p)}(X)$, since fundamental classes are compatible with ρ; the class b itself may depend on the choice of Z. In particular, $(f_*^{\Lambda_{(p)}}1)^p - f_*^{\Lambda_{(p)}}1 \in p\Lambda^{0,0}_{(p)}(X)$.

Recall from Section 3 that the class of a stack such as Y is computed on a good cover, a finite flat cover by a derived scheme of rank prime to p, and then divided by the rank of the cover. This is the role of Z and r in the theorem.

4.1. **Outline.**

The count for finite sets. If S is a finite set with d elements, the cyclic group C_p acts on S^p by rotating the factors. It fixes the d constant tuples, and it acts freely on the other $d^p - d$ tuples because p is prime. Hence

$$d^p = d + p \cdot \#\{\text{non-constant necklaces in } S\},$$

where a necklace is a C_p-orbit. We want the same decomposition of the class $(f_*1)^p = [W^p \to X]$. The constant maps should contribute f_*1, and the non-constant ones should form a free quotient and contribute p times a class.

Why W^p itself does not work. For a finite syntomic W this decomposition cannot be carried out on W^p. The complement of the diagonal is not proper over X, so it has no fundamental class. Its natural compactification, the blowup $\mathrm{Bl}_\Delta W^p$ of the diagonal, is proper, but C_p has fixed points on the exceptional divisor. Indeed, a p-group acting linearly on a nonzero vector space in characteristic p fixes a nonzero vector, and so it fixes a point of the projectivized normal space of the diagonal. Since C_p acts trivially on the diagonal itself, such a point is a fixed point on $\mathrm{Bl}_\Delta W^p$.

The case $p = 2$ and $W = \mathbb{A}^n$ shows this concretely. The swap of $\mathbb{A}^n \times \mathbb{A}^n$ acts on the normal direction $x_1 - x_0$ of the diagonal by -1, which equals 1 in characteristic 2. So the swap fixes the whole exceptional divisor of $\mathrm{Bl}_\Delta(\mathbb{A}^n \times \mathbb{A}^n)$, and the quotient by the swap is not a free quotient there.

Deforming the group. The remedy is to let the group itself vary over the blowup. For $p = 2$ consider the group schemes $G_q = \{z^2 = qz\}$ inside the additive group, depending on a parameter q. For $q \neq 0$ the group G_q has the two points $z = 0$ and $z = q$, so it is C_2. For $q = 0$ it is $\alpha_2 = \{z^2 = 0\}$. A map from G_q to $\mathbb{A}^n$ is a polynomial $v(z) = v_0 + b_1 z$, determined by its values $x_0 = v_0$ and $x_1 = v_0 + qb_1$ when $q \neq 0$. Translation $z \mapsto z + \xi$ by a point ξ of G_q changes the map to $v(z + \xi) = (v_0 + \xi b_1) + b_1 z$. For $\xi = q$ this swaps x_0 and x_1, since $x_0 + qb_1 = x_1$ and $x_1 + qb_1 = x_0 + 2qb_1 = x_0$ in characteristic 2. For $q = 0$ it is the action $v_0 \mapsto v_0 + \xi b_1$ of α_2, which is free as long as $b_1 \neq 0$.

Now consider the triples (v_0, b_1, q) with $b_1 \in \mathbb{A}^n \setminus 0$ (a vector, not a scalar), up to the scaling $(v_0, b_1, q) \mapsto (v_0, ab_1, a^{-1}q)$, which does not change the values x_0, x_1. The invariants are v_0, the difference $x_1 - x_0 = qb_1$ and the line $[b_1]$, so the quotient is $\mathrm{Bl}_\Delta(\mathbb{A}^n \times \mathbb{A}^n)$. Its exceptional divisor is $q = 0$, and there $[b_1]$ records the normal direction along which $x_1 - x_0$ tends to zero. On the quotient, q is no longer a function but a section of a line bundle, and the group G_q, twisted by this line bundle (the group scheme $G_{q,\mathcal{L}}$ of Section 4.5), acts on the blowup. Off the exceptional divisor it acts by the swap, and on the exceptional divisor it acts freely through α_2. So the fixed points of the swap have been replaced by a free action of a different group of order 2.

To compare this blowup with W^p we add a deformation parameter t. For $p = 2$ and $W = \mathbb{A}^n$ the space $\mathfrak{B}$ we use is the blowup of $W^2 \times \mathbb{A}^1_t$ along $\Delta \times 0$. Its fibre over $t \neq 0$ is W^2. Its fibre over $t = 0$ has two components. One is the strict transform $D \simeq \mathrm{Bl}_\Delta W^2$ of $W^2 \times 0$, on which G_q acts freely. The other is the exceptional divisor E over $\Delta \times 0$, which consists of the maps from α_2 to W, constant or not.

The deformation for general p. In general $G_q = \{z^p = q^{p-1}z\}$ is a family of group schemes of order p with $G_q \simeq C_p$ for $q \neq 0$ and $G_0 = \alpha_p = \{z^p = 0\}$. A map from G_q to $\mathbb{A}^n$ is a polynomial $v(z) = v_0 + \sum_{j=1}^{p-1} b_j z^j$. For $q \neq 0$ it is the p-tuple of its values at the points of G_q, and for $q = 0$ it is a jet of order $p-1$. Translation by α_p acts freely on the non-constant jets. Indeed, if b_m is the top nonzero coefficient, then translation by ξ changes the coefficient of z^{m-1} by $mb_m\xi$, and m is invertible since $m < p$.

The weights of the blowup are forced by the scaling. Replacing (q, z) by $(a^{-1}q, a^{-1}z)$ preserves the equation $z^p = q^{p-1}z$, and it preserves the map v only if b_j is replaced by $a^j b_j$. So b_j has weight j, and dividing the locus $b \neq 0$ by this scaling gives a weighted projective space with weights $1, \dots, p-1$. For $W = \mathbb{A}^n$ the resulting space $\mathfrak{B}$ is a weighted blowup of $W^p \times \mathbb{A}^1_t$, and for general W it is a derived closed substack of one. It is a stack, but a tame one, because the weights are prime to p, so its stabilizers are groups μ_m with $m < p$. Tameness is what allows us to compute its classes on good covers of rank prime to p.

The three identities. The space $\mathfrak{B}$ (Section 4.2) is proper over $W^p \times \mathbb{P}^1$, with fibre W^p at infinity. Its fibre over 0 is the zero locus of $t = qT$, where q is a section of a line bundle $\mathcal{L}$ and T of $\mathcal{O}_{\mathbb{P}^1}(1) \otimes \mathcal{L}^{-1}$. Since first Chern classes add in an additive theory, this fibre is the sum of the two boundary divisors $E = \{q = 0\}$ and $D = \{T = 0\}$. In terms of the normalized classes of Section 3 we prove

$$(4.2) \quad (f_*1)^p = \langle E \to X\rangle + \langle D \to X\rangle, \qquad \langle E \to X\rangle = f_*1, \qquad \langle D \to X\rangle = p\langle Y \to X\rangle.$$

Subtracting the second identity from the first and using the third gives $(f_*1)^p - f_*1 = p\langle Y \to X\rangle$, which is the theorem. The three identities correspond to the three terms of the necklace count.

(i) The first identity compares the fibres of $\mathfrak{B}$ over 0 and over infinity (Lemma 4.2). It is the analogue of the decomposition of S^p into constant and non-constant tuples.

(ii) The divisor E consists of the maps from α_p to W. After a linear deformation it becomes the zero locus of a section of a vector bundle on a weighted projective bundle over W, and its class is f_*1 (Lemma 4.3). This is the term d.

(iii) On D the twisted group $G_{q,\mathcal{L}}$ of Section 4.5 acts freely, with quotient the stack Y. The quotient map $D \to Y$ is a torsor under a finite flat group scheme of rank p, so its trace is p by Lemma 4.6, and the third identity follows (Lemma 4.7). This is the term $p \cdot \#\{\text{necklaces}\}$.

When X is a point and W consists of d reduced points, everything is classical and the identities are the count for finite sets. The divisor E is W, embedded as the constant maps, because a jet in a reduced zero-dimensional scheme is constant. The divisor D is $W^p \setminus \Delta$, and Y is the set of non-constant necklaces.

We may assume that X is connected, so that f has constant degree d, and that $d \geq 1$. Since W is finite over the affine scheme X, we may choose a closed immersion $W \hookrightarrow \mathbb{A}^n_X$ with $n \geq 1$. Classes with $\mathrm{MGL}_{(p)}$ coefficients are used only for $Z \to X$ in the construction of b; all unmarked fundamental classes below have coefficients in $\Lambda_{(p)}$, and all identities are identities in $\Lambda_{(p)}$.

4.2. **A weighted deformation.** We now construct the space $\mathfrak{B}$ of the outline for general p and W. It is built in three steps: the family of groups G_q, the space R_q of maps from G_q to W, and the quotient of a modification of R_q by the scaling of q. The point of the construction is that $\mathfrak{B}$ is proper over $W^p \times \mathbb{P}^1$, so that its fibres over 0 and ∞ have fundamental classes that can be compared.

The family of groups. We need a finite flat family of group schemes of order p that is C_p over $q \neq 0$ and α_p over $q = 0$. Such a family is given by the Rees construction, which in general degenerates a filtered algebra to its associated graded algebra. The Hopf algebra of $C_p = \operatorname{Spec} \mathcal{O}_X[z_0]/(z_0^p - z_0)$ is filtered by polynomial degree in z_0, and the comultiplication $z_0 \mapsto z_0 \otimes 1 + 1 \otimes z_0$ preserves the filtration. The Rees algebra is generated over $\mathcal{O}_X[q]$ by

$z = qz_0$, and the relation $z_0^p = z_0$ becomes $z^p = q^p z_0 = q^{p-1}z$. Hence we obtain the finite flat group scheme

$$G_q = \operatorname{Spec} \mathcal{O}_X[q,z]/(z^p - q^{p-1}z) \tag{4.3}$$

of rank p over $X \times \mathbb{A}^1_q$, a closed subgroup of the additive group $\mathbf{G}_a \times \mathbb{A}^1_q$. For $q \neq 0$ the group G_q is constant, with the p points $z = iq$ for $i \in \mathbb{F}_p$, and the isomorphism $G_q \simeq C_p$ sends iq to i. Its fibre at $q = 0$ is the associated graded group $\alpha_p = \operatorname{Spec} \mathcal{O}_X[z]/(z^p)$.

The space of maps. Let R_q be the space of maps from G_q to W over X. Thus an S-point of R_q, for S over $X \times \mathbb{A}^1_q$, is a map $G_q \times_{X\times\mathbb{A}^1_q} S \to W$ over X. For $q \neq 0$ such a map is a p-tuple of points of W, its values at the points iq, so the fibre of R_q over $q \neq 0$ is W^p. For $q = 0$ it is a map from α_p, that is, a jet of order $p-1$ in W.

To see the structure of R_q we first take $W = \mathbb{A}^n_X$. Since $\mathcal{O}[q,z]/(z^p - q^{p-1}z)$ is free over $\mathcal{O}[q]$ with basis $1, z, \dots, z^{p-1}$, a map $G_q \to \mathbb{A}^n$ is a polynomial

$$v(z) = v_0 + \sum_{j=1}^{p-1} b_j z^j, \qquad v_0, b_j \in \mathbb{A}^n,$$

and the space of these maps is the affine space $\mathbb{A}^{np}$ over $X \times \mathbb{A}^1_q$ with coordinates v_0 and $b = (b_1, \dots, b_{p-1})$. For general W the condition that v lands in W is the system of equations $g(v(z)) = 0$ in $\mathcal{O}[z]/(z^p - q^{p-1}z)$, where g runs over the equations of W in $\mathbb{A}^n_X$. Expanding in the basis z^j, each g gives p equations in the coordinates v_0, b. These equations need not form a regular sequence, even when W is a complete intersection, and we keep track of this by taking R_q to be the derived scheme they cut out. Formally, R_q is the derived Weil restriction

$$R_q = \operatorname{Res}_{G_q/(X\times\mathbb{A}^1_q)}(W \times_X G_q).$$

We use the following two facts, which are Lemma 1.11 applied to the finite locally free map $G_q \to X \times \mathbb{A}^1_q$ of rank p.

(i) Since $W \times_X G_q$ is quasi-smooth of virtual dimension zero over G_q, the derived scheme R_q is quasi-smooth of virtual dimension zero over $X \times \mathbb{A}^1_q$, by the remark after Lemma 1.11.
(ii) The closed immersion $W \hookrightarrow \mathbb{A}^n_X$ induces a closed immersion of R_q into the Weil restriction of $\mathbb{A}^n$, which is the affine space of polynomials v above.

Over $q \neq 0$ the derived scheme R_q is W^p, since there G_q is a disjoint union of p copies of the base.

The scaling and the weights. The family R_q is not proper over $\mathbb{A}^1_q$: as $q \to 0$, the p values of v come together, and the coefficients b_j can tend to infinity. We compactify by dividing by a scaling, as in the case $p = 2$ of the outline. Let $\mathbb{G}_m$ act on (q, z) with weight -1, so that a sends (q, z) to $(a^{-1}q, a^{-1}z)$. This preserves the equation $z^p = q^{p-1}z$. For the polynomial v to be preserved, $b_j z^j$ must have weight zero, which forces b_j to have weight j, while v_0 has weight zero. Thus $\mathbb{G}_m$ acts on R_q. To compare with the constant family W^p we need a second coordinate T such that $t = qT$ is invariant. This forces T to have weight 1, and T plays the role of q^{-1}. Set

$$\mathfrak{B} = [(R_q \times \mathbb{A}^1_T \setminus \{b = T = 0\})/\mathbb{G}_m], \qquad t = qT.$$

The function t has weight zero, so it is a function on $\mathfrak{B}$. On the locus $T \neq 0$ we may normalize $T = 1$, and we recover R_q with $q = t$. Over $q = 0$ the quotient of the nonzero vectors (T, b) by $\mathbb{G}_m$ is a weighted projective space with weights 1 on T and j on the coordinates of b_j. The locus $b = T = 0$ must be removed. Over $q = 0$ its points would have stabilizer $\mathbb{G}_m$, and over $q \neq 0$ it is, in the model case below, exactly the centre of the blowup. The following computation shows that after its removal $\mathfrak{B}$ is proper over $W^p \times \mathbb{A}^1_t$.

The model case. For $W = \mathbb{A}^n_X$ the stack $\mathfrak{B}$ is a weighted blowup. Indeed, the invariant coordinates $y_j = q^j b_j$ and t identify $\mathfrak{B}$ with the weighted blowup of $\mathbb{A}^n_{v_0} \times \mathbb{A}^{n(p-1)}_y \times \mathbb{A}^1_t$ along $y = t = 0$, with weights j on y_j and 1 on t. Evaluating v at the points iq of G_q gives the Vandermonde change of coordinates

$$x_i = v(iq) = v_0 + \sum_{j=1}^{p-1} i^j y_j \quad (i \in \mathbb{F}_p),$$

which identifies (v_0, y) with the p points (x_i) of $(\mathbb{A}^n)^p$. The centre $y = t = 0$ is $\Delta \times 0$. For $p = 2$ this is the ordinary blowup of $(\mathbb{A}^n)^2 \times \mathbb{A}^1_t$ along $\Delta \times 0$ described in the outline.

For general W, the equations saying that v lands in W cut out $\mathfrak{B}$ as a derived closed substack of this weighted blowup. It is proper over $W^p \times \mathbb{A}^1_t$: the map to $(\mathbb{A}^n)^p \times \mathbb{A}^1_t$ evaluates v at the sections iq of G_q, and since v lands in W so do the values $x_i = v(iq)$; hence the proper map from $\mathfrak{B}$ factors through the closed substack $W^p \times \mathbb{A}^1_t$. Over $t \neq 0$ it is $W^p \times \mathbb{G}_m$. Gluing this open part to $W^p \times (\mathbb{P}^1 \setminus 0)$, we obtain a stack, again denoted $\mathfrak{B}$, with a proper map $\mathfrak{B} \to W^p \times \mathbb{P}^1_t$ whose fibre at infinity is W^p. Choose homogeneous coordinates $[s_0 : s_\infty]$ on $\mathbb{P}^1$, viewed as sections of $\mathcal{O}_{\mathbb{P}^1}(1)$, with $t = s_0/s_\infty$ on the original affine chart. Thus s_0 vanishes at 0 and s_∞ at infinity.

Tameness and a good cover. To compute fundamental classes on $\mathfrak{B}$ we need it to be tame and to have a good cover by a derived scheme. Tameness holds because the stabilizers of $\mathfrak{B}$ have order prime to p. For $q \neq 0$ the stabilizers are trivial, since q has weight -1. At $q = 0$ some coordinate among T, b_j is nonzero, and these coordinates have weights in $\{1, \dots, p-1\}$, so the stabilizers are subgroups of μ_m with $m < p$. Thus $\mathfrak{B}$ is a tame Deligne–Mumford stack. It is quasi-smooth of virtual dimension one over X, since $R_q \times \mathbb{A}^1_T$ has virtual dimension two over X and $\mathbb{G}_m$ has dimension one. On the original quotient chart, let $\mathcal{L}$ be the line bundle of character -1. There q is a section of $\mathcal{L}$, and the coordinates of b_j are sections of $\mathcal{L}^{-j}$. On the overlap $t \neq 0$, the section q is nowhere vanishing. Glue $\mathcal{L}$ to the trivial line on $W^p \times (\mathbb{P}^1 \setminus 0)$ using this q-trivialization. This extends the line bundle and its section to the compactification, with $q = 1$ on the new chart. The coefficients b_j also extend as sections of $\mathcal{L}^{-j}$: in this trivialization they are the invariant functions $y_j = q^j b_j$ on W^p. We retain this notation for the extended line bundle and sections.

The good cover is obtained by extracting roots, which turns the weighted blowup into an ordinary one. Put

$$N = n(p-1), \qquad A = ((p-1)!)^n.$$

Let $Z_{\mathfrak{B}} \to \mathfrak{B}$ be obtained by adjoining sections $U_{j,\ell}$ of $\mathcal{L}^{-1}$ with $U^j_{j,\ell} = b_{j,\ell}$, where $b_{j,\ell}$, for $1 \leq \ell \leq n$, are the coordinates of b_j. By the root-cover calculation of Section 3, $Z_{\mathfrak{B}} \to \mathfrak{B}$ is a good cover of rank $\prod_j j^n = A$. Its source is a derived scheme. Indeed, the invariant coordinates $u_{j,\ell} = qU_{j,\ell}$ satisfy $y_{j,\ell} = u^j_{j,\ell}$. Since T and all $U_{j,\ell}$ now have weight 1, the ambient weighted blowup becomes the ordinary blowup

$$\mathcal{P} = \mathrm{Bl}_{\{u=t=0\}}\big(\mathbb{A}^n_{v_0} \times \mathbb{A}^N_u \times \mathbb{P}^1_t\big)_X.$$

Thus $Z_{\mathfrak{B}}$ is a derived closed subscheme of the smooth scheme $\mathcal{P}$, proper over X. For $k \gg 0$ the line bundle $\mathcal{L}^{-1} \otimes \mathcal{O}_{\mathbb{P}^1}(k)$ restricts to an ample bundle on $t_0 Z_{\mathfrak{B}}$, because $\mathcal{L}^{-1}$ restricts to $\mathcal{O}(-\mathrm{Exc})$, which is relatively ample for the blowup, and $\mathcal{O}_{\mathbb{P}^1}(1)$ is ample on the base $\mathbb{A}^{n+N}_X \times \mathbb{P}^1$, X being affine. So the fundamental classes of Theorem 1.6 are defined for $Z_{\mathfrak{B}}$ and its quasi-smooth closed subschemes.

4.3. **The two boundary classes.** On the original affine chart, the scaling coordinate T is a section of $\mathcal{L}^{-1}$. Multiplying it by s_∞ gives a section of $\mathcal{O}_{\mathbb{P}^1}(1) \otimes \mathcal{L}^{-1}$. On the other chart, where $q = 1$, use the section s_0. These sections agree on the overlap because $qT = t = s_0/s_\infty$. They define a global section, again denoted T, satisfying $qT = s_0$. Pullbacks of the line bundles and sections from $\mathbb{P}^1$ are suppressed in this notation.

The derived zero loci

$$E = V(q), \qquad D = V(T)$$

are proper and quasi-smooth of virtual dimension zero over X. For these derived virtual Cartier divisors we write

$$\mathcal{O}(E) := \mathcal{L}, \qquad \mathcal{O}(D) := \mathcal{O}_{\mathbb{P}^1}(1) \otimes \mathcal{L}^{-1}$$

for their defining line bundles. The fibre over 0 is the derived zero locus of the product $qT = s_0$. The divisors E, D are the two kinds of degenerate limits of p-tuples of points of W.

- The divisor E lies over $q = 0$, so it consists of maps from α_p to W, that is, of jets of order $p-1$ in W, up to scaling. It contains the constant jets, and it will contribute the term f_*1. We compute its class as an Euler class in Lemma 4.3.
- The divisor D is the locus $T = 0$. Since the locus $b = T = 0$ was removed, it consists of the non-constant maps $G_q \to W$ (those with $b \neq 0$) up to scaling. The twisted group $G_{q,\mathcal{L}}$ acts freely on it, and this is the source of the factor p (Section 4.5). For nonreduced W it need not be the classical strict transform of $W^p \times 0$.

Over $t \neq 0$ the cover $Z_{\mathfrak{B}}$ is cut out by $u^j_{j,\ell} = y_{j,\ell}$ in the invariant coordinates, which do not involve t. It therefore extends over $W^p \times (\mathbb{P}^1 \setminus 0)$ as the product of $\mathbb{P}^1 \setminus 0$ with a finite flat cover of W^p of rank A. Let Z_E, Z_D, Z_∞ be the restrictions of $Z_{\mathfrak{B}}$ to E, D and the fibre W^p at ∞. These are good covers of rank A, and they carry the restrictions of the ample line bundle on $t_0 Z_{\mathfrak{B}}$. The first identity of (4.2) says that the fibre of $\mathfrak{B}$ over ∞ has the same class as the sum of the two components of its fibre over 0.

Lemma 4.2. $(f_*1)^p = \langle E \to X\rangle + \langle D \to X\rangle$.

Proof. The idea is that the fibres of $\mathfrak{B}$ over 0 and over ∞ in $\mathbb{P}^1_t$ are zero loci of sections of the same line bundle $\mathcal{O}_{\mathbb{P}^1}(1)$, and the fibre over 0 splits as a sum because $s_0 = qT$. We compute on the good cover $Z_{\mathfrak{B}}$, pulling back the defining line bundles. The sections q and T multiply to the section s_0 of $\mathcal{O}_{\mathbb{P}^1}(1)$, which gives

$$\mathcal{O}(E) \otimes \mathcal{O}(D) \simeq \mathcal{O}_{\mathbb{P}^1}(1) \tag{4.4}$$

on $Z_{\mathfrak{B}}$. The formal group law of $\Lambda_{(p)}$ is additive, so the first Chern classes of the two sides agree: $c_1(\mathcal{O}(E)) + c_1(\mathcal{O}(D)) = c_1(\mathcal{O}_{\mathbb{P}^1}(1))$.

The zero loci of q, T and s_∞ are Z_E, Z_D and Z_∞. By the Euler class rule Corollary 1.7(iv), their fundamental classes are obtained by capping the respective first Chern classes with $[Z_{\mathfrak{B}}/X]$ and pushing forward along $g\colon Z_{\mathfrak{B}} \to X$. The identity of first Chern classes therefore gives

$$[Z_E \to X] + [Z_D \to X] = [Z_\infty \to X].$$

The map $Z_\infty \to W^p$ is a good cover of rank A, so the composition and product rules of Corollary 1.7 give $[Z_\infty \to X] = A[W^p \to X] = A(f_*1)^p$. Dividing by A and using the definition (3.1) of normalized classes proves the lemma. □

4.4. **Linearizing the exceptional divisor.** We now show that the class of E is f_*1, which is the term d of the necklace count. The divisor E lies over $q = 0$, where G_q is α_p, with algebra $\mathcal{O}[z]/(z^p)$. A map $\alpha_p \to W$ is a jet in W, and the equations expressing that a jet lies in W are not linear in the b_j. For example, if $p \geq 3$ and g is an equation of W, the coefficient of z^2 in $g(v_0 + b_1 z + b_2 z^2 + \cdots)$ involves the second derivatives of g applied to b_1. So E is not visibly the zero locus of a section of a vector bundle, and we cannot compute its class by the Euler class formula.

The square-zero deformation. The equations become linear after deforming $\mathcal{O}[z]/(z^p)$ to an algebra whose augmentation ideal has square zero, because maps into a square-zero extension are derivations. To keep the weights we deform within graded algebras. Consider

$$\mathcal{B}_\lambda = \mathcal{O} \oplus \bigoplus_{j=1}^{p-1} \mathcal{O} e_j, \qquad e_i e_j = \begin{cases} \lambda e_{i+j}, & i+j < p, \\ 0, & i+j \geq p, \end{cases}$$

with e_j of weight $-j$ and λ of weight zero. This is a finite free commutative algebra over $\mathcal{O}[\lambda]$. At $\lambda = 1$ it is $\mathcal{O}[z]/(z^p)$, with $e_j = z^j$, and at $\lambda = 0$ its augmentation ideal has square zero.

Replacing $R_q|_{q=0}$ in the definition of E by the derived Weil restriction along $\mathcal{B}_\lambda$, we obtain a quasi-smooth family

$$E_\lambda \subseteq \mathbb{A}^n_{v_0} \times \mathbb{P}(1, 1^n, 2^n, \dots, (p-1)^n) \times \mathbb{A}^1_\lambda$$

of virtual dimension zero over $X \times \mathbb{A}^1_\lambda$, by Lemma 1.11 applied to the finite locally free map $\operatorname{Spec} \mathcal{B}_\lambda \to X \times \mathbb{A}^1_\lambda$ of rank p. The homogeneous coordinates are T, of weight 1, and the coordinates of the b_j, of weight j. The family is proper: its equations are $\mathbb{G}_m$-homogeneous for every λ, because $\mathcal{B}_\lambda$ is graded with λ of weight 0; the unstable locus $b = T = 0$ is removed; evaluation at the augmentation sends E_λ to the finite X-scheme W; and the weighted projective space is proper. Extracting the same roots $b_{j,\ell} = U_{j,\ell}^j$ gives a good cover $Z_{E,\lambda} \to E_\lambda$ of rank A inside $\mathbb{A}^n_{v_0} \times \mathbb{P}^N \times \mathbb{A}^1_\lambda$, with homogeneous coordinates T and $U_{j,\ell}$ on $\mathbb{P}^N$. Its fibre at $\lambda = 1$ is Z_E, since $E = \big[(\operatorname{Res}_{\alpha_p}(W) \times \mathbb{A}^1_T \setminus \{b = T = 0\})/\mathbb{G}_m\big]$ is the same construction for $\mathcal{B}_1 = \mathcal{O}[z]/(z^p)$.

The linear fibre. At $\lambda = 0$ the family E_0 is the zero locus of a section of a vector bundle on a weighted projective bundle over W. Let $\mathcal{N}$ be the conormal bundle of $W \hookrightarrow \mathbb{A}^n_X$, of rank n since W is finite over X. If W is locally cut out by $g_1, \dots, g_n$, then $g(v_0 + \sum_j b_j e_j) = g(v_0) + \sum_j dg(v_0) b_j\, e_j$ in $\mathcal{B}_0$, since all products of the e_j vanish. So the equations of E_0 are $g(v_0) = 0$, that is $v_0 \in W$, and $dg(v_0) b_j = 0$ for $1 \leq j \leq p-1$. In other words, E_0 is the derived zero locus of a section of

$$\mathcal{V} = \bigoplus_{j=1}^{p-1} \mathcal{N}^\vee \otimes \mathcal{O}(j)$$

on the weighted projective bundle $\mathbb{P}(1, 1^n, \dots, (p-1)^n)$ over W. This description does not depend on the equations g_i, since derivations are represented by the cotangent complex $\mathbb{L}_{W/X} = [\mathcal{N} \to \Omega_{\mathbb{A}^n_X/X}|_W]$, with $\Omega_{\mathbb{A}^n_X/X}|_W = \mathcal{O}_W^n$, and $dg\colon \mathcal{O}_W^n \to \mathcal{N}^\vee$ is dual to $\mathcal{N} \to \Omega_{\mathbb{A}^n_X/X}|_W$.

The class of E_0 is now computed by the projective bundle formula. The Euler class of $\mathcal{V}$ is a polynomial in $h = c_1(\mathcal{O}(1))$ whose leading coefficient is the product of the weights j^n, which is A, and pushing forward to W picks out exactly this coefficient.

Lemma 4.3. $\langle E \to X \rangle = f_* 1$.

Proof. We first move to $\lambda = 0$. The family $Z_{E,\lambda}$ over $X \times \mathbb{A}^1_\lambda$ has fibres $Z_{E,1} = Z_E$ and $Z_{E,0}$. By derived base change their classes are the pullbacks of $[Z_{E,\lambda} \to X \times \mathbb{A}^1]$ along $\lambda = 1$ and $\lambda = 0$, and these pullbacks agree by homotopy invariance. Hence $[Z_E \to X] = [Z_{E,0} \to X]$.

The cover $Z_{E,0}$ is the zero locus of the pullback of $\mathcal{V}$ to the root cover $\mathbb{P}^N_W$. Write $\pi\colon \mathbb{P}^N_W \to W$ and $h = c_1(\mathcal{O}(1))$. The root map pulls $\mathcal{O}(j)$ back to $\mathcal{O}(j)$; we use $\mathcal{V}$ also for the pulled-back bundle. Since $\mathcal{N}^\vee$ has rank n, the additive formal group law gives

$$c_N(\mathcal{V}) = \prod_{j=1}^{p-1} c_n(\mathcal{N}^\vee \otimes \mathcal{O}(j)) = A h^N + \textstyle\sum_{0 \leq i < N} \pi^*(\gamma_i) h^i,$$

with $\gamma_i \in \Lambda^{2(N-i), N-i}_{(p)}(W)$, since $c_n(\mathcal{N}^\vee \otimes \mathcal{O}(j))$ is $j^n h^n$ plus terms of lower degree in h. By Lemma 1.9, $\pi_* h^N = 1$ and $\pi_* h^i = 0$ for $i < N$.

Since $\pi\colon \mathbb{P}^N_W \to W$ is smooth, the Euler class rule Corollary 1.7(iv) over the base W gives $[Z_{E,0} \to W] = \pi_* c_N(\mathcal{V}) = A$. By the composition rule of Corollary 1.7, $[Z_E \to X] = [Z_{E,0} \to X] = Af_*1$. Dividing by A proves the lemma. □

4.5. **The free boundary and its quotient.** It remains to show that the class of D is p times a class. In the count for finite sets this is the statement that C_p acts freely on the non-constant tuples. We show that G_q acts freely on the non-constant maps, so that the twisted group $G_{q,\mathcal{L}}$ acts freely on D, form the quotient Y, and check that Y is a stack whose class is defined.

The action. The group G_q acts on maps $G_q \to W$ by translating the source. A point ξ of G_q sends the polynomial $v(z)$ to $v'(z) = v(z+\xi)$, that is,

$$b'_j = \sum_{k=j}^{p-1} \binom{k}{j} b_k \xi^{k-j}, \qquad v'_0 = v_0 + \sum_{k=1}^{p-1} b_k \xi^k, \qquad \xi^p = q^{p-1}\xi. \tag{4.5}$$

Here $v(z+\xi)$ is computed in $\mathcal{O}[z]/(z^p - q^{p-1}z)$. The substitution $z \mapsto z+\xi$ is well defined because G_q is a subgroup of the additive group: in characteristic p we have $(z+\xi)^p - q^{p-1}(z+\xi) = (z^p - q^{p-1}z) + (\xi^p - q^{p-1}\xi)$. For $q \neq 0$ and $\xi = q$ this is the rotation $x_i \mapsto x_{i+1}$ of the values. For $q = 0$ it is the action of α_p on jets. For $p = 2$ it is the action $v_0 \mapsto v_0 + \xi b_1$ of the outline.

The constant maps $G_q \to W$ are those with $b = 0$. They form the closed subscheme $W \times \mathbb{A}^1_q$ of R_q, the image of the section given by constant maps, and every point of it is fixed by G_q. Let U be the open subscheme $b \neq 0$ of the ambient polynomial space. Then $R_q \cap U$ is the space of *non-constant* maps $G_q \to W$, and $D = [((R_q \cap U) \times \{T = 0\})/\mathbb{G}_m] = [(R_q \cap U)/\mathbb{G}_m]$, since D is the locus $T = 0$ and the point $b = T = 0$ was removed in the definition of $\mathfrak{B}$. Freeness on U is the analogue of the fact that C_p acts freely on non-constant p-tuples.

Lemma 4.4. *The action of G_q on U is free.*

Proof. The stabilizer group scheme of the action is finite over U. Its augmentation ideal is a finite $\mathcal{O}_U$-module, so by Nakayama's lemma it suffices to show that the stabilizers at geometric points are trivial. For $q \neq 0$ the action is rotation of a non-constant p-tuple, whose stabilizer in $\mathbb{Z}/p$ is trivial since p is prime. For $q = 0$, let m be the largest index with $b_m \neq 0$. By (4.5), translation by ξ changes b_{m-1} by $mb_m\xi$ if $m \geq 2$, and it changes v_0 by $b_1\xi$ if $m = 1$. Since $1 \leq m < p$, the integer m is invertible, so the stabilizer is the subscheme $\xi = 0$. □

The quotient. Since the action is free, we may pass to the quotient. The map $U \to V = U/G_q$ is a finite flat G_q-torsor, and V is a smooth algebraic space over $X \times \mathbb{A}^1_q$, because flatness and regularity of the fibres descend along the finite flat cover $U \to V$. The G_q-invariant subscheme $R_q \cap U$ descends to a quasi-smooth closed subspace of V. Scaling acts on the base and on the translation parameter by $(q, \xi) \mapsto (a^{-1}q, a^{-1}\xi)$. The translation action is $\mathbb{G}_m$-equivariant for these actions, so scaling descends to V and to $(R_q \cap U)/G_q$. We may therefore form the successive quotient

$$Y = [((R_q \cap U)/G_q)/\mathbb{G}_m] \subseteq [V/\mathbb{G}_m].$$

This description shows that Y is quasi-smooth of virtual dimension zero over X, as D is. The line bundle $\mathcal{L}$ and its section q descend to Y. The group acting on the fibres of $D \to Y$ is not G_q itself, because q is now a section of $\mathcal{L}$ rather than a function. It is the twist of G_q by $\mathcal{L}$,

$$G_{q,\mathcal{L}} = \ker(F - q^{p-1}\colon \mathcal{L} \to \mathcal{L}^p),$$

a finite flat group scheme of rank p over Y, where F is the p-th power map. Thus $D \to Y$ is a $G_{q,\mathcal{L}}$-torsor.

The stack Y is the stack of non-constant necklaces in W. Its open substack $q \neq 0$ is, after normalizing $q = 1$ by the $\mathbb{G}_m$-action, the quotient of $W^p \setminus \Delta$ by rotation. Its closed substack

$q = 0$ is the stack of non-constant jets of order $p-1$ in W, modulo translation by α_p and rescaling of the source.

The class of Y is defined. For the class $\langle Y \to X \rangle$ to make sense we need a good cover of Y by a derived scheme with an ample line bundle. The general existence result Proposition 3.5 supplies one once Y is tame, carries a faithful line bundle, and has a projective coarse space. The next lemma verifies these hypotheses.

Lemma 4.5. *The stack Y is separated, proper and tame Deligne–Mumford over X. Its stabilizers are subgroups of μ_m for some $m < p$, and $\mathcal{L}$ is faithful. Moreover Y has a good cover $Z \to Y$ of prime-to-p rank r, with an ample line bundle on $t_0 Z$.*

Proof. We first compute the stabilizers. A stabilizer of the successive quotient Y maps to $\mathbb{G}_m$ by remembering the scaling factor. Its kernel is a stabilizer of the translation action, which is trivial by Lemma 4.4; hence this map is injective. For $q \neq 0$ its image is trivial, since q has weight -1. For $q = 0$, the highest nonzero coefficient b_m is unchanged by translation, so the scaling factor a of a stabilizing element satisfies $a^m = 1$. Hence the stabilizers are subgroups of μ_m with $m < p$. This proves that Y is tame Deligne–Mumford, and $\mathcal{L}$ is faithful because μ_m acts on it through the inclusion $\mu_m \subseteq \mathbb{G}_m$.

Separatedness and properness descend from D along the finite flat surjection $D \to Y$. Here D is of finite type and separated over X, being closed in the part of $\mathfrak{B}$ over $\mathbb{A}^1_t$, a closed substack of a weighted blowup, and Y is of finite type since $D \to Y$ is a finite flat cover. The pullback of the diagonal of Y along $D \times_X D \to Y \times_X Y$ is finite, so Y is separated, and Y is proper since D is proper and $D \to Y$ is surjective.

It remains to construct the good cover, which we do by checking the hypotheses of Proposition 3.5 with $\mathcal{M} = \mathcal{L}^{-1}$. We have already shown that Y is separated, tame and carries the faithful line bundle $\mathcal{L}$. The sheaves $\pi_j \mathcal{O}_Y$ are coherent and vanish for $j \gg 0$, since Y is quasi-smooth over X.

The ampleness hypothesis is checked on D and descended. The cover Z_D is closed in the base change of $\mathrm{Bl}_0(\mathbb{A}^N_u)$ to the finite X-scheme $\widetilde{W^p} = W^p \times_{\mathbb{A}^N_y} \mathbb{A}^N_u$ (here $W^p \to \mathbb{A}^N_y$ records the coordinates $y_{j,\ell}$ of a p-tuple through the Vandermonde change of coordinates, and $\mathbb{A}^N_u \to \mathbb{A}^N_y$ is $u_{j,\ell} \mapsto u^j_{j,\ell}$), and $\mathcal{L}^{-1}$ restricts to the relatively ample bundle $\mathcal{O}(1)$ of this blowup. Hence $\mathcal{L}^{-1}$ is ample on $t_0 Z_D$. With $c = (p-1)!$, the bundle $\mathcal{L}^{-c}$ descends to the coarse spaces D_c and Y_c. The maps $t_0 Z_D \to D_c \to Y_c$ are finite surjective, the second being induced by the representable finite surjection $D \to Y$. Now we use the following fact [SP, Tag 0GFB]: if $f \colon Y' \to Y''$ is a finite surjective morphism of algebraic spaces proper over a Noetherian ring and $\mathcal{N}$ is a line bundle on Y'' whose pullback to Y' is ample, then $\mathcal{N}$ is ample, and in particular Y'' is a scheme. Applied to $t_0 Z_D \to Y_c$, it shows that Y_c is a scheme on which the descended bundle is ample. In particular Y_c is projective over X. Thus Proposition 3.5 applies and gives the good cover $Z \to Y$. □

4.6. **The trace of the quotient torsor.** To compare the classes of D and Y we need the trace of the torsor $D \to Y$. In the count for finite sets each necklace is the image of exactly p tuples. Here the corresponding statement is that a torsor under a finite flat group scheme of rank p has trace p in $\Lambda_{(p)}$. This requires an argument, just as the trace condition in the definition of good covers had to be checked for root covers. For the groups $G_{q,\mathcal{L}}$ it holds because the torsor is cut out by a monic equation of degree p, so that Corollary 1.10 applies. We state it over a derived scheme and apply it after base change to a good cover of Y.

Lemma 4.6. *Let S be a derived scheme, $\mathcal{L}$ a line bundle, $a \in \Gamma(S, \mathcal{L}^{p-1})$, and $P \to S$ a torsor under $\ker(F - a \colon \mathcal{L} \to \mathcal{L}^p)$. Then $[P \to S]_{\Lambda_{(p)}} = p$, compatibly with derived base change.*

Proof. The idea is to realize P as a divisor of degree p in a $\mathbb{P}^1$-bundle that avoids infinity, and to apply Corollary 1.10. Write $G = \ker(F - a)$. The group G acts on $\mathcal{O} \oplus \mathcal{L}$ by

$(x, y) \mapsto (x, y + \xi x)$. Twisting by P gives an extension $0 \to \mathcal{L} \to V \to \mathcal{O} \to 0$, and in $\mathbb{P}(V)$ the complement of $\mathbb{P}(\mathcal{L})$ is the affine bundle $P \times^G \mathcal{L}$. The polynomial $y^p - ayx^{p-1}$ is G-invariant, since $(y + \xi x)^p - a(y + \xi x)x^{p-1} = y^p - ayx^{p-1} + (\xi^p - a\xi)x^p$. It therefore defines a section of $\mathcal{O}(p) \otimes \pi^*\mathcal{L}^p$ on $\mathbb{P}(V)$. This section does not vanish on $\mathbb{P}(\mathcal{L})$, where $x = 0$ and $y^p \neq 0$. On the affine bundle its zero locus is $P \times^G G \simeq P$. Thus Corollary 1.10 gives $[P \to S] = p$, also after any derived base change. □

The third identity of (4.2) now follows by computing both classes on compatible good covers.

Lemma 4.7. $\langle D \to X\rangle = p\langle Y \to X\rangle$.

Proof. We pull the good cover of Y back to D. Choose the good cover $Z \to Y$ of rank r supplied by Lemma 4.5. Then $D_Z = D \times_Y Z$ is finite over the derived scheme Z, hence is a derived scheme, and $D_Z \to D$ is a good cover of rank r. It carries the pullback of the ample bundle on t_0Z, which is ample since $D_Z \to Z$ is finite. The map $D_Z \to Z$ is a torsor under $G_{q,\mathcal{L}}$, so its trace is p by Lemma 4.6. In Lemma 4.2 the class $\langle D \to X\rangle$ was computed on Z_D; by Lemma 3.3 we may compute it on D_Z instead, and the composition rule of Corollary 1.7 gives

$$\langle D \to X\rangle = r^{-1}[D_Z \to X] = p\, r^{-1}[Z \to X] = p\langle Y \to X\rangle.$$

□

4.7. **Proof of the theorem.** The three identities identify $\langle Y \to X\rangle$ as a class whose product with p is $(f_*1)^p - f_*1$. To obtain (4.1), we lift this specified class to $\mathrm{MGL}_{(p)}$ using the same scheme cover of Y.

Proof of Theorem 4.1. Substituting Lemmas 4.3 and 4.7 into Lemma 4.2 gives

$$(f_*^{\Lambda_{(p)}}1)^p - f_*^{\Lambda_{(p)}}1 = p\langle Y \to X\rangle = p\, r^{-1}[Z \to X]_{\Lambda_{(p)}}.$$

To lift this to cobordism, note that $Z \to X$ is proper and quasi-smooth, and smoothable by Lemma 1.8, since Z carries a line bundle ample on t_0Z. Hence $Z \to X$ also has a fundamental class in $\mathrm{MGL}_{(p)}$. Set $b = r^{-1}[Z \to X]_{\mathrm{MGL}_{(p)}}$. Fundamental classes are compatible with the map ρ of MGL-algebras, so $\rho(u) = f_*^{\Lambda_{(p)}}1$ and $\rho(b) = r^{-1}[Z \to X]_{\Lambda_{(p)}}$. This proves (4.1). □

5. From finite syntomic classes to the comparison theorem

In this section we prove that $\Phi\colon \Lambda_{(p)} \to H\mathbb{Z}_{(p)}$ is an equivalence over $\mathbb{F}_p$ (Theorem 5.11). Equivalently, the fibre

$$F = \mathrm{fib}\big(\Phi\colon \Lambda_{(p)} \longrightarrow H\mathbb{Z}_{(p)}\big) \tag{5.1}$$

vanishes. The geometric results of the previous sections enter only through one statement about cobordism, Proposition 5.2. The rest of the argument is homotopy theory.

The logic of the proof. Suppose that $F \neq 0$. Both $\Lambda_{(p)}$ and $H\mathbb{Z}_{(p)}$ are effective, and for effective spectra the homotopy t-structure can be tested on the weight-zero homotopy sheaves $\underline{\pi}_{i,0}$ alone (Proposition 5.3). Hence F has a first nonzero weight-zero homotopy sheaf $\underline{\pi}_{s,0}F$, and $s \geq 1$ by Lemma 5.7. Such a sheaf is detected at generic points (Theorem 5.4), so $\pi_{s,0}F(K) \neq 0$ for some finitely generated field $K/\mathbb{F}_p$. Since $H\mathbb{Z}_{(p)}$ has no weight-zero homotopy in positive degrees, this group equals $\pi_{s,0}\Lambda_{(p)}(K)$. We then show that $\pi_{s,0}\Lambda_{(p)}(K) = 0$, a contradiction, by combining three facts.

(i) *The group $\pi_{s,0}\Lambda_{(p)}(K)$ is p-primary torsion.* This follows from Hoyois's theorem that Φ becomes an equivalence once p is inverted.

(ii) *Every p-primary torsion element of $\pi_{s,0}\Lambda_{(p)}(K)$ lifts to $\pi_{s,0}\mathrm{MGL}_{(p)}(K)$* (Lemma 5.10). Here the minimality of s is used. The fibre of $\rho\colon \mathrm{MGL}_{(p)} \to \Lambda_{(p)}$ is filtered by Lazard weight, and a Bockstein argument using the theorem of Geisser and Levine shows that the obstruction to lifting kills p-primary torsion.

(iii) *The image $I_s(K)$ of $\pi_{s,0}\mathrm{MGL}_{(p)}(K)$ in $\pi_{s,0}\Lambda_{(p)}(K)$ is zero* (Proposition 5.2). This is where the geometry enters. The factorial bound of Section 2 shows that $I_s(K)$ has bounded exponent, and the necklace formula of Section 4 shows that $I_s(K)$ is p-divisible.

By (i) and (ii) every element of $\pi_{s,0}\Lambda_{(p)}(K)$ lies in $I_s(K)$, which is zero by (iii). We prove (iii) first, since it only needs the results of the previous sections, and then set up the homotopy theory for (i) and (ii).

5.1. **The image of cobordism.** Let $I_s(K)$ be the image of ρ_*, as in Proposition 2.8. By Proposition 2.8, $s!\, I_s(K) = 0$. To show that $I_s(K) = 0$ it therefore suffices to show that $I_s(K)$ is p-divisible, since a p-divisible $\mathbb{Z}_{(p)}$-module of bounded exponent is zero.

Divisibility comes from the necklace formula of Theorem 4.1 and the following lemma.

Lemma 5.1. *Let R be a commutative motivic ring spectrum, K a field, $s > 0$, and $P \in \mathbb{Z}[t]$. Let $\sigma\colon S^{s,0}_K \to \mathcal{Y}$ be a pointed map of motivic spaces and $u \in R^{0,0}(\mathcal{Y})$. Write $\sigma^* u = a + v$ with a constant and v reduced. Then $\sigma^* P(u) = P(a) + P'(a)v$.*

Here $R^{0,0}(S^{s,0}_K) = R^{0,0}(K) \oplus \widetilde{R}^{0,0}(S^{s,0}_K)$, where the first summand consists of the constant classes, pulled back from the base point, and the second of the reduced classes, which vanish at the base point. The reduced summand is $\pi_{s,0}R(K)$.

Proof. For $s > 0$ the sphere $S^{s,0}_K$ is a suspension. Hence its reduced diagonal is null, and the product of two reduced classes vanishes. In particular $v^2 = 0$. Since $R^{0,0}(S^{s,0}_K)$ is commutative, Taylor expansion of P at a gives $P(a + v) = P(a) + P'(a)v$. □

Proposition 5.2. *$I_s(K) = 0$ for every finitely generated field $K/\mathbb{F}_p$ and every $s > 0$.*

Proof. The idea is to apply the necklace formula to the universal family over a Hilbert scheme, pull it back along a map from a sphere that represents a given cobordism class, and compare reduced parts. By Proposition 2.8, $p^e I_s(K) = 0$ with $e = v_p(s!)$. Hence it suffices to show $I_s(K) = pI_s(K)$, for then $I_s(K) = p^e I_s(K) = 0$.

Elements of $\pi_{s,0}\mathrm{MGL}_{(p)}(K)$ are fractions x/a with $x \in \pi_{s,0}\mathrm{MGL}(K)$ and $p \nmid a$. Since a is a unit in $\mathbb{Z}_{(p)}$, it suffices to show $\rho(x) \in pI_s(K)$ for $x \in \pi_{s,0}\mathrm{MGL}(K)$.

We first represent x geometrically, using the Hilbert scheme model Theorem 2.7(2) over K. Let $H = \mathrm{Hilb}^{\mathrm{lci}}_d(\mathbb{A}^m)$, a smooth $\mathbb{F}_p$-scheme, with universal family $f\colon W \to H$. Since spheres are compact, there are a finite stage H and a pointed map $\sigma\colon S^{s,0}_K \to H_K$ in the pointed motivic homotopy category over K such that x is the pullback along σ of $f_*1 - d$. Put $u = f_*^{\mathrm{MGL}_{(p)}} 1 \in \mathrm{MGL}^{0,0}_{(p)}(H)$.

Next we apply the necklace formula to f. The scheme H need not be affine, whereas Theorem 4.1 is stated over smooth affine bases. Let $\pi\colon J \to H$ be a Jouanolou torsor over $\mathbb{F}_p$, so that J is smooth affine and π is an affine bundle, hence an $\mathbb{A}^1$-equivalence. Applying Theorem 4.1 on J to the base change of f, whose class is $\pi^* u$, gives $b_J \in \mathrm{MGL}^{0,0}_{(p)}(J)$ with $\rho(\pi^* u^p - \pi^* u) = p\rho(b_J)$. As π^* is an isomorphism on $\mathrm{MGL}^{0,0}_{(p)}$ and on $\Lambda^{0,0}_{(p)}$, compatible with ρ, we have $b_J = \pi^* b$ for some $b \in \mathrm{MGL}^{0,0}_{(p)}(H)$, and

$$\rho(u^p - u) = p\rho(b).$$

We pull this identity back along $H_K \to H$ and keep the notation u, b for the restrictions.

Finally, we take reduced parts along σ. Write $\sigma^* u = d + v$ and $\sigma^* b = b_0 + y$ with v, y reduced. The constant part of $\sigma^* u$ is d, since $\pi_{0,0}\mathrm{MGL}_{(p)}(K) = \mathbb{Z}_{(p)}$ (Lemma 5.7) and f

has degree d, and the reduced part is $v = x$. By Lemma 5.1, applied to $R = \Lambda_{(p)}$ and $P(t) = t^p - t$, the reduced part of $\rho(u^p - u) = p\rho(b)$ is

$$(pd^{p-1} - 1)\rho(v) = p\rho(y).$$

Here $\rho(y)$ lies in $I_s(K)$, because b is a cobordism class and not merely a class in $\Lambda_{(p)}$. Moreover $pd^{p-1} - 1$ is a unit in $\mathbb{Z}_{(p)}$. Hence $\rho(x) \in pI_s(K)$. □

5.2. **The homotopy-theoretic inputs.** We now collect the facts from motivic homotopy theory needed for steps (i) and (ii) of the outline. Throughout, k denotes a perfect field; in the applications $k = \mathbb{F}_p$.

The homotopy t-structure. The induction runs over the first nonzero homotopy sheaf of F, so we need a notion of connectivity for motivic spectra. With the homotopy modules $h_i(E)_m = \underline{\pi}_{i-m,-m}E$ of the Conventions, the homotopy t-structure on $\mathrm{SH}(k)$ is defined by declaring $E \in \mathrm{SH}(k)_{\geq r}$ if and only if $h_i(E) = 0$ for all $i < r$, that is, $\underline{\pi}_{i-m,-m}E = 0$ for all $i < r$ and all m. We say that E is *r-connective* if $E \in \mathrm{SH}(k)_{\geq r}$, and *connective* if it is 0-connective. A suspension shifts the homotopy modules as follows: for $n \in \mathbb{Z}$,

$$h_i(\Sigma^{2n,n}E)_m = h_{i-n}(E)_{m+n}. \tag{5.2}$$

Indeed, both sides are $\underline{\pi}_{i-m-2n,-m-n}E$. Since the weight index m runs over all integers, (5.2) says that $\Sigma^{2n,n}$ maps $\mathrm{SH}(k)_{\geq r}$ to $\mathrm{SH}(k)_{\geq r+n}$. For $n > 0$ this is the sense in which a Tate twist moves homotopy into higher connectivity, equivalently, in which the homotopy of E in degree s only sees h_{s-n} of the twisted piece.

Testing connectivity involves all weights m. For effective spectra, that is, for the objects of the localizing subcategory $\mathrm{SH}(k)^{\mathrm{eff}}$ generated by the suspension spectra $\Sigma^\infty_+ X$ of smooth k-schemes, weight zero suffices.

Proposition 5.3 ([B2, Proposition 4])**.** *Let k be perfect and $E \in \mathrm{SH}(k)^{\mathrm{eff}}$. Then $E \in \mathrm{SH}(k)^{\mathrm{eff}}_{\geq r}$ if and only if $\underline{\pi}_{i,0}E = 0$ for $i < r$. Moreover $\mathrm{SH}(k)^{\mathrm{eff}}_{\geq r} \subset \mathrm{SH}(k)_{\geq r}$.*

Here $\mathrm{SH}(k)^{\mathrm{eff}}_{\geq r}$ is the non-negative part of the effective homotopy t-structure, generated under colimits and extensions by the S^1-suspensions $\Sigma^r\Sigma^\infty_+ X$, X smooth.

In particular, if E is effective and all $\underline{\pi}_{i,0}E$ vanish, then $E \in \mathrm{SH}(k)_{\geq r}$ for every r, so $E = 0$ since the homotopy t-structure is non-degenerate. Moreover, if $\underline{\pi}_{i,0}E = 0$ for $i < s$ then E is s-connective, so that $h_i(E) = 0$ in every weight for $i < s$. This is the form in which we use the proposition. We also use that effective spectra and connective spectra are each closed under colimits, extensions, and $\Sigma^{2n,n}$ for $n \geq 0$. For connective spectra and $\Sigma^{2n,n}$ this is (5.2).

Effectivity of $\Lambda_{(p)}$ and $H\mathbb{Z}_{(p)}$. Proposition 5.3 applies to F because both $\Lambda_{(p)}$ and $H\mathbb{Z}_{(p)}$ are effective. For $\Lambda_{(p)}$ this holds because MGL is effective and $\Lambda_{(p)}$ is the colimit of the spectra $\Lambda_i = \mathrm{MGL}_{(p)}/(a_1, \ldots, a_i)$, where Λ_i is the cofibre of $a_i \colon \Sigma^{2i,i}\Lambda_{i-1} \to \Lambda_{i-1}$ for $i > 0$, and these are maps between effective spectra. For motivic cohomology, effectivity follows from the identification of $H\mathbb{Z}$ with the zero slice $s_0\mathbf{1}$ of the sphere spectrum [H, Remark 4.20], [FS, Theorem 6.12], since the zero slice of an effective spectrum is effective. In weight zero, motivic cohomology has the homotopy sheaves of a point:

$$\underline{\pi}_{i,0}H\mathbb{Z}_{(p)} = 0 \ (i \neq 0), \qquad \underline{\pi}_{0,0}H\mathbb{Z}_{(p)} = \mathbb{Z}_{(p)}. \tag{5.3}$$

On a field K the group $\pi_{i,0}H\mathbb{Z}(K)$ is the motivic cohomology group $H^{-i,0}(K, \mathbb{Z})$ of weight zero, which is $\mathbb{Z}$ for $i = 0$ and vanishes otherwise.

Homotopy sheaves are detected at generic points. To pass from the vanishing of a homotopy sheaf to a statement about fields we use Morel's theory of strictly $\mathbb{A}^1$-invariant sheaves. A Nisnevich sheaf of abelian groups M on smooth k-schemes is strictly $\mathbb{A}^1$-invariant if $H^n_{\mathrm{Nis}}(X, M) \to H^n_{\mathrm{Nis}}(X \times \mathbb{A}^1, M)$ is an isomorphism for all smooth X and all $n \geq 0$. The homotopy sheaves $\underline{\pi}_{a,b}E$ of every $E \in \mathrm{SH}(k)$ are strictly $\mathbb{A}^1$-invariant.

Theorem 5.4 ([H, Theorem 2.7])**.** *Let k be a perfect field and $f\colon M \to N$ a morphism of strictly $\mathbb{A}^1$-invariant Nisnevich sheaves on smooth k-schemes. Then f is an isomorphism if and only if $f(K)\colon M(K) \to N(K)$ is an isomorphism for every finitely generated field extension K/k.*

Applied to $M \to 0$, it shows that a strictly $\mathbb{A}^1$-invariant sheaf M is zero if and only if $M(K) = 0$ for every finitely generated field K/k. This is how the theorem is used below; it rests on Morel's result that such sheaves are unramified.

Two theorems on motivic cohomology. The first theorem controls $H\mathbb{Z}/p$ in characteristic p. Geisser and Levine proved that for a field K of characteristic p the motivic cohomology $H^{a,b}(K, \mathbb{Z}/p)$ vanishes for $a \neq b$, and is the group of logarithmic de Rham–Witt forms $\nu^b(K)$ for $a = b$. In terms of homotopy modules, $h_i(H\mathbb{Z}/p)_m(K) = \pi_{i-m,-m}(H\mathbb{Z}/p)(K) = H^{m-i,m}(K, \mathbb{Z}/p)$ vanishes for $i \neq 0$. Since homotopy sheaves are detected on fields by Theorem 5.4, this is the following statement.

Theorem 5.5 (Geisser–Levine [GL, Theorems 1.1 and 8.3])**.** *Over $\mathbb{F}_p$, the spectrum $H\mathbb{Z}/p$ lies in the heart of the homotopy t-structure, that is, $h_i(H\mathbb{Z}/p) = 0$ for $i \neq 0$.*

Here $H\mathbb{Z}/p$ is the cofibre of p on $H\mathbb{Z}$; it equals $H\mathbb{Z}_{(p)}/p$, the target of $\Lambda_{(p)}/p \to H\mathbb{Z}/p$ in Lemma 5.10.

The second theorem is the comparison with p inverted, the statement of Theorem A away from the characteristic.

Theorem 5.6 (Hoyois [H, Theorem 7.12 and Remark 7.15])**.** *Over a field of characteristic exponent c, the map $\Phi_k\colon \mathrm{MGL}/(a_1, a_2, \ldots) \to H\mathbb{Z}$ becomes an equivalence after inverting c. In particular, over $\mathbb{F}_p$ the map $\Phi[1/p]\colon \Lambda_{(p)}[1/p] \to H\mathbb{Z}_{(p)}[1/p]$ is an equivalence.*

We use Theorem 5.6 in the following form. Let $K/\mathbb{F}_p$ be a finitely generated field and $s > 0$. Since Φ is an isomorphism on homotopy after inverting p, and $\pi_{s,0}H\mathbb{Z}_{(p)}(K) = 0$ by (5.3), we get $\pi_{s,0}\Lambda_{(p)}(K)[1/p] = 0$. Here we use $\pi_{s,0}(\Lambda_{(p)}[1/p])(K) = \pi_{s,0}\Lambda_{(p)}(K)[1/p]$, which holds because $\Lambda_{(p)}[1/p]$ is the filtered colimit of $\Lambda_{(p)} \xrightarrow{p} \Lambda_{(p)} \xrightarrow{p} \cdots$ and the functor $\pi_{s,0}(-)(K)$ commutes with filtered colimits. Every element of $\pi_{s,0}\Lambda_{(p)}(K)$ is therefore killed by a power of p, that is,

$$\pi_{s,0}\Lambda_{(p)}(K) \text{ is a } p\text{-primary torsion group.} \tag{5.4}$$

The bottom of the fibre. The first input is that F has nothing in degrees ≤ 0, so that the first nonzero homotopy sheaf, if any, sits in a positive degree s, where (5.4) and Proposition 5.2 apply.

Lemma 5.7. *The fibre F is effective and 1-connective, and $\underline{\pi}_{0,0}\Lambda_{(p)} = \mathbb{Z}_{(p)}$ as sheaves of rings.*

Proof. We first compute $\underline{\pi}_{0,0}\Lambda_{(p)}$ and then read off the low homotopy of F from the long exact sequence. By [BEHKSWY, Proposition 4.1 and Theorem 1.1], the sheaf $\underline{\pi}_{0,0}\mathrm{MGL}$ is the constant sheaf $\mathbb{Z}$, and the class f_*1 of a finite syntomic morphism f of degree d corresponds to d. Passing from Λ_{i-1} to Λ_i takes the cofibre of $a_i\colon \Sigma^{2i,i}\Lambda_{i-1} \to \Lambda_{i-1}$. The spectrum Λ_{i-1} is connective, being built from $\mathrm{MGL}_{(p)}$ by cofibres of maps from connective spectra. Hence the source $\Sigma^{2i,i}\Lambda_{i-1}$ is i-connective by (5.2), with $i \geq 1$, so its h_0 and h_{-1} vanish and the cofibre map $\Lambda_{i-1} \to \Lambda_i$ induces an isomorphism on h_0. Since h_0 commutes with filtered colimits, we obtain $\underline{\pi}_{0,0}\Lambda_{(p)} = \mathbb{Z}_{(p)}$.

The map Φ preserves units, so it is an isomorphism $\mathbb{Z}_{(p)} \to \mathbb{Z}_{(p)}$ on $\underline{\pi}_{0,0}$. Both $\Lambda_{(p)}$ and $H\mathbb{Z}_{(p)}$ are connective, and $\underline{\pi}_{1,0}H\mathbb{Z}_{(p)} = 0$ by (5.3). The long exact sequence

$$\underline{\pi}_{1,0}H\mathbb{Z}_{(p)} \to \underline{\pi}_{0,0}F \to \underline{\pi}_{0,0}\Lambda_{(p)} \xrightarrow{\cong} \underline{\pi}_{0,0}H\mathbb{Z}_{(p)} \to \underline{\pi}_{-1,0}F \to \underline{\pi}_{-1,0}\Lambda_{(p)} = 0$$

of $F \to \Lambda_{(p)} \to H\mathbb{Z}_{(p)}$, together with the vanishing of $\underline{\pi}_{i,0}$ of $\Lambda_{(p)}$ and $H\mathbb{Z}_{(p)}$ for $i < 0$, shows that $\underline{\pi}_{i,0}F = 0$ for $i \leq 0$. The fibre F is effective because $\Lambda_{(p)}$ and $H\mathbb{Z}_{(p)}$ are, and Proposition 5.3 shows that it is 1-connective. □

5.3. **The filtration by Lazard weight.**

The problem. Step (ii) of the outline asks us to lift a class in $\pi_{s,0}\Lambda_{(p)}(K)$ along $\rho\colon \mathrm{MGL}_{(p)} \to \Lambda_{(p)}$. The obstruction lies in $\pi_{s-1,0}\,\mathrm{fib}(\rho)(K)$, so we must understand the fibre of ρ in homotopy degree $s-1$. We want to describe this fibre in terms of $\Lambda_{(p)}$ itself, because the minimality of s gives information about $\Lambda_{(p)}$ in low degrees and nothing directly about $\mathrm{MGL}_{(p)}$.

The idea. In algebra the kernel of the augmentation $\epsilon\colon L_{(p)} \to \mathbb{Z}_{(p)}$ is the ideal of elements of positive weight, where a_i has weight i. Filtering $L_{(p)}$ by the ideals $L_{\geq n}$ of elements of weight at least n, one sees that each graded piece $L_{\geq n}/L_{\geq n+1}$ is a module over $L_{(p)}$ on which every a_i acts by zero. It is therefore a module over $L_{(p)}/(a_1, a_2, \ldots) = \mathbb{Z}_{(p)}$, namely a free $\mathbb{Z}_{(p)}$-module placed in weight n. The commutative model (1.1) turns this into a filtration of $\mathrm{MGL}_{(p)}$ itself, because it is a base change functor that can be applied to any graded $L_{(p)}$-module. A graded piece that is a free $\mathbb{Z}_{(p)}$-module in weight n becomes a sum of copies of $\Lambda_{(p)}$ shifted by $\Sigma^{2n,n}$, since the realization functor $e_{\mathbb{S}}$ sends weight n to $(\Sigma^{2,1}\mathbf{1})^{\otimes n}$. By (5.2), these shifted copies are n-connective, so only the pieces with $n < s$ affect the homotopy that matters for the lifting problem.

Let $L_{\geq n} \subset L_{(p)}$ be the ideal of elements of weight at least n, and put $L_n = L_{\geq n}/L_{\geq n+1}$. These are graded $L_{(p)}$-modules. In the direct sums below, m ranges over monomials in the fixed generators a_i of $L_{(p)}$, with $\mathrm{wt}(a_i) = i$. For each $n \geq 0$ there are finitely many such monomials of weight n; for $n = 0$ the sole monomial is 1.

Definition 5.8. For $n \geq 0$ put

$$\mathcal{F}^n = \mathrm{MGL}_{(p)} \otimes_{\mathcal{A}} e_{\mathbb{S}}U(L_{\geq n}),$$

with $\mathcal{A} = e_{\mathbb{S}}U(L_{(p)})$ as in Definition 1.2.

Lemma 5.9. *We have $\mathcal{F}^0 = \mathrm{MGL}_{(p)}$, $\mathcal{F}^1 = \mathrm{fib}(\rho)$, and cofibre sequences*

$$\mathcal{F}^{n+1} \longrightarrow \mathcal{F}^n \longrightarrow \bigoplus_{\mathrm{wt}(m)=n} \Sigma^{2n,n}\Lambda_{(p)}$$

for $n \geq 0$. Each $\mathcal{F}^n$ is effective and n-connective.

Proof. The functor $N \mapsto \mathrm{MGL}_{(p)} \otimes_{\mathcal{A}} e_{\mathbb{S}}U(N)$ on graded $L_{(p)}$-modules preserves cofibre sequences, and it sends $L_{(p)}$ to $\mathrm{MGL}_{(p)} \otimes_{\mathcal{A}} \mathcal{A} = \mathrm{MGL}_{(p)}$, so $\mathcal{F}^0 = \mathrm{MGL}_{(p)}$. Applied to the exact sequence $L_{\geq 1} \to L_{(p)} \xrightarrow{\epsilon} \mathbb{Z}_{(p)}$ it gives the fibre sequence $\mathcal{F}^1 \to \mathrm{MGL}_{(p)} \xrightarrow{\rho} \Lambda_{(p)}$, by the definition (1.1) of $\Lambda_{(p)}$. Hence $\mathcal{F}^1 = \mathrm{fib}(\rho)$. Applied to $L_{\geq n+1} \to L_{\geq n} \to L_n$ it gives a cofibre sequence $\mathcal{F}^{n+1} \to \mathcal{F}^n \to \mathrm{MGL}_{(p)} \otimes_{\mathcal{A}} e_{\mathbb{S}}U(L_n)$.

We identify the third term. The graded module L_n is concentrated in weight n, with basis the monomials of that weight. Elements of positive weight map $L_{\geq n}$ into $L_{\geq n+1}$, so they act by zero on L_n. Its $L_{(p)}$-action therefore factors through the augmentation, and as a graded $L_{(p)}$-module it is

$$L_n \simeq \bigoplus_{\mathrm{wt}(m)=n} \mathbb{Z}_{(p)}(n),$$

where (n) denotes the grading shift of (1.2). Applying the exact module functor of Proposition 1.3(ii) gives

$$\mathrm{MGL}_{(p)} \otimes_{\mathcal{A}} e_{\mathbb{S}} U(L_n) \simeq \bigoplus_{\mathrm{wt}(m)=n} \Sigma^{2n,n}\Lambda_{(p)}.$$

For connectivity, choose a graded free resolution of $L_{\geq n}$ concentrated in nonnegative homological degrees, with every generator of weight at least n. Such a resolution is constructed recursively: $L_{\geq n}$ has no components below weight n, a graded free module on generators of weights at least n surjects onto it, and the kernel again has no components below weight n. The same construction therefore applies at every step of the resolution.

By (1.2), the exact module functor sends a free generator of weight w to $\Sigma^{2w,w}\mathrm{MGL}_{(p)}$. Applying it to the resolution expresses $\mathcal{F}^n$ as a colimit built from cells

$$\Sigma^{2w,w}\mathrm{MGL}_{(p)}[j], \qquad w \geq n, \quad j \geq 0.$$

Here the resolution is recovered from its successive finite skeleta, whose quotients are the terms of the resolution in their homological degrees. Each displayed cell is effective, since $w \geq 0$, and is $(w+j)$-connective by (5.2) and the connectivity of $\mathrm{MGL}_{(p)}$. Closure under extensions and colimits now shows that $\mathcal{F}^n$ is effective and n-connective. □

5.4. **Lifting torsion classes.** We now carry out step (ii) of the outline. Let $s \geq 1$ and suppose that F is s-connective. Consider $x \in \pi_{s,0}\Lambda_{(p)}(K)$. The fibre sequence $\mathcal{F}^1 \to \mathrm{MGL}_{(p)} \xrightarrow{\rho} \Lambda_{(p)}$ of Lemma 5.9 gives an exact sequence

$$\pi_{s,0}\mathrm{MGL}_{(p)}(K) \xrightarrow{\rho_*} \pi_{s,0}\Lambda_{(p)}(K) \xrightarrow{\partial} h_{s-1}(\mathcal{F}^1)_0(K),$$

so x lifts if and only if $\partial x = 0$. If x is killed by p^r, then so is ∂x. It therefore suffices to show that the group $h_{s-1}(\mathcal{F}^1)_0(K)$ has no p-torsion.

In words: only the finite sums of $\Sigma^{2n,n}\Lambda_{(p)}$ with $1 \leq n < s$ contribute to h_{s-1}, and modulo p their homotopy in degree s comes from $h_j(\Lambda_{(p)}/p)$ with $1 \leq j < s$, which injects into $h_j(H\mathbb{Z}/p) = 0$. The Bockstein sequence then rules out p-torsion in degree $s-1$.

Lemma 5.10. *Suppose that F is s-connective, where $s \geq 1$. Then for every essentially smooth field $K/\mathbb{F}_p$, every p-primary torsion element of $\pi_{s,0}\Lambda_{(p)}(K)$ lifts to $\pi_{s,0}\mathrm{MGL}_{(p)}(K)$.*

Proof. By the discussion above it suffices to show that multiplication by p is injective on $h_{s-1}(\mathcal{F}^1)_0(K)$.

We first discard the part of the filtration above s. Put $Q_s = \mathrm{cofib}(\mathcal{F}^s \to \mathcal{F}^1)$, so that $Q_1 = 0$. Since $\mathcal{F}^s$ is s-connective by Lemma 5.9, the map $\mathcal{F}^1 \to Q_s$ induces an isomorphism $h_{s-1}(\mathcal{F}^1) \cong h_{s-1}(Q_s)$. By the same lemma, Q_s has a finite filtration with graded pieces

$$\bigoplus_{\mathrm{wt}(m)=n} \Sigma^{2n,n}\Lambda_{(p)}, \qquad 1 \leq n < s.$$

Next we compute modulo p. Since F is s-connective, so is F/p. The long exact sequence of $F/p \to \Lambda_{(p)}/p \to H\mathbb{Z}/p$ shows that $h_j(\Lambda_{(p)}/p)$ injects into $h_j(H\mathbb{Z}/p)$ for $j < s$, and the latter vanishes for $j \neq 0$ by Theorem 5.5. Hence $h_j(\Lambda_{(p)}/p) = 0$ for $1 \leq j < s$. By (5.2) we obtain

$$h_s(\Sigma^{2n,n}(\Lambda_{(p)}/p))_m = h_{s-n}(\Lambda_{(p)}/p)_{m+n} = 0 \qquad (1 \leq n < s),$$

since $1 \leq s-n < s$. Each weight has finitely many monomials, so the graded pieces of Q_s/p are finite sums of $\Sigma^{2n,n}(\Lambda_{(p)}/p)$, and therefore $h_s(Q_s/p) = 0$ by the finite filtration.

The cofibre sequence $Q_s \xrightarrow{p} Q_s \to Q_s/p$ gives an exact sequence

$$h_s(Q_s/p) \longrightarrow h_{s-1}(Q_s) \xrightarrow{p} h_{s-1}(Q_s).$$

Since the first term vanishes, multiplication by p is injective on $h_{s-1}(Q_s) \cong h_{s-1}(\mathcal{F}^1)$, and hence on its sections over K. Finally, in the exact sequence

$$\pi_{s,0}\mathrm{MGL}_{(p)}(K) \longrightarrow \pi_{s,0}\Lambda_{(p)}(K) \xrightarrow{\partial} h_{s-1}(\mathcal{F}^1)_0(K)$$

the target of ∂ has no p-torsion. Hence ∂ kills every p-primary torsion element, and such an element lifts to $\pi_{s,0}\mathrm{MGL}_{(p)}(K)$. □

5.5. **The comparison.** We now assemble the three steps of the outline.

Theorem 5.11. *Over $\mathbb{F}_p$, the map $\Phi\colon \Lambda_{(p)} \to H\mathbb{Z}_{(p)}$ is an equivalence.*

Proof. We show that the fibre F of (5.1) has no first nonzero homotopy sheaf. Suppose $F \neq 0$. Since F is effective, Proposition 5.3 shows that some $\underline{\pi}_{i,0}F$ is nonzero, and by Lemma 5.7 this happens only for $i \geq 1$. Let $s \geq 1$ be the least such i. Then F is s-connective by Proposition 5.3.

The sheaf $\underline{\pi}_{s,0}F$ is strictly $\mathbb{A}^1$-invariant and nonzero, so by Theorem 5.4 we have $\pi_{s,0}F(K) \neq 0$ for some finitely generated field $K/\mathbb{F}_p$. By (5.3) we have $\pi_{s,0}H\mathbb{Z}_{(p)}(K) = \pi_{s+1,0}H\mathbb{Z}_{(p)}(K) = 0$, so the long exact sequence of $F \to \Lambda_{(p)} \to H\mathbb{Z}_{(p)}$ gives $\pi_{s,0}F(K) \cong \pi_{s,0}\Lambda_{(p)}(K)$, so $\pi_{s,0}\Lambda_{(p)}(K) \neq 0$.

This group is p-primary torsion by (5.4). Since F is s-connective, Lemma 5.10 shows that every element of it lies in the image $I_s(K)$ of $\pi_{s,0}\mathrm{MGL}_{(p)}(K)$. But $I_s(K) = 0$ by Proposition 5.2. This contradicts $\pi_{s,0}\Lambda_{(p)}(K) \neq 0$. □

6. Arbitrary fields, $\operatorname{Spec}\mathbb{Z}_p$ and completion

We deduce Theorem A from Theorem 5.11. Over a field k of characteristic p two facts are needed: both sides of Φ_k are pulled back from $\mathbb{F}_p$ after p-localization, and the prime-to-p part is Theorem 5.6. Over $\operatorname{Spec}\mathbb{Z}_p$ the map is checked on the two points, using Spitzweck's identification of the restrictions of $M\mathbb{Z}$.

Theorem 6.1. *For every field k of characteristic p, the map $\Phi_k\colon \mathrm{MGL}/(a_1, a_2, \ldots) \to H\mathbb{Z}$ is an equivalence.*

Proof. We reduce to $\mathbb{F}_p$ by base change and treat the primes different from p by Hoyois's theorem. Let $C = \operatorname{cofib}(\Phi_k)$ and let $\pi\colon \operatorname{Spec} k \to \operatorname{Spec}\mathbb{F}_p$ be the structure morphism. The functor π^* preserves colimits and is symmetric monoidal, and $\pi^*\mathrm{MGL} = \mathrm{MGL}$. Hence it preserves the iterated cofibres of the a_i, so $\pi^*(\mathrm{MGL}/(a_1, a_2, \ldots)) = \mathrm{MGL}/(a_1, a_2, \ldots)$. Motivic cohomology is also preserved: since $\mathbb{F}_p$ is perfect, k is a filtered colimit of smooth $\mathbb{F}_p$-algebras, and $H\mathbb{Z}$ is stable under essentially smooth base change [H, Theorem 4.18], [S, Lemma 8.23]. The map Φ_k is constructed as in Lemma 1.4: that argument only uses the vanishing of motivic cohomology of the base in negative weight, which holds over any field, so it gives, integrally and over k, an MGL-module map Φ_k extending the unit $\mathrm{MGL} \to H\mathbb{Z}$, unique up to a contractible choice; the same holds p-locally. As π^* is symmetric monoidal and preserves units, $\pi^*\Phi$ is such an extension over k, so $\pi^*\Phi = (\Phi_k)_{(p)}$ by uniqueness. Hence $C_{(p)} = \pi^* \operatorname{cofib}(\Phi)$, which vanishes by Theorem 5.11. Moreover $C[1/p] = 0$ by Theorem 5.6.

It remains to see that these two facts give $C = 0$. Let G be one of the compact generators $\Sigma^{a,b}\Sigma^\infty_+ X$ of $\mathrm{SH}(k)$, with X smooth over k, and let $x\colon G \to C$ be a map. Since G is compact, $[G, C_{(p)}] = [G, C] \otimes \mathbb{Z}_{(p)}$ and $[G, C[1/p]] = [G, C] \otimes \mathbb{Z}[1/p]$. Both groups vanish, so the element x is annihilated by an integer prime to p and by a power of p. Hence $x = 0$. As this holds for all compact generators, $C = 0$. □

Over $\operatorname{Spec}\mathbb{Z}_p$ we use Spitzweck's motivic cohomology spectrum $M\mathbb{Z}$ [S]. It is oriented with the additive formal group law [S, Proposition 10.1 and Remark 10.2], and the map $\Phi_{\mathbb{Z}_p}$ of Theorem A is the MGL-module map extending the unit $\mathrm{MGL} \to M\mathbb{Z}$, constructed as in Lemma 1.4. Existence and uniqueness up to a contractible choice hold because the argument only uses that $M\mathbb{Z}^{*,b}(\operatorname{Spec}\mathbb{Z}_p) = 0$ for $b < 0$ [S]. We use the following identification of its restrictions to the two points of $\operatorname{Spec}\mathbb{Z}_p$. Let $j\colon \operatorname{Spec}\mathbb{Q}_p \to \operatorname{Spec}\mathbb{Z}_p$ and $i\colon \operatorname{Spec}\mathbb{F}_p \to \operatorname{Spec}\mathbb{Z}_p$ be the inclusions of the generic and the closed point.

Theorem 6.2 (Spitzweck [S, Theorems 7.18, 8.22 and 8.25])**.** *There are equivalences $j^*M\mathbb{Z} \simeq H\mathbb{Z}$ in $\mathrm{SH}(\mathbb{Q}_p)$ and $i^*M\mathbb{Z} \simeq H\mathbb{Z}$ in $\mathrm{SH}(\mathbb{F}_p)$, compatible with the orientations.*

Theorem 6.3. *The map* $\Phi_{\mathbb{Z}_p}\colon \mathrm{MGL}/(a_1, a_2, \ldots) \to M\mathbb{Z}$ *is an equivalence in* $\mathrm{SH}(\operatorname{Spec}\mathbb{Z}_p)$.

Proof. We check the equivalence on the generic and the closed point. Let $C = \operatorname{cofib}(\Phi_{\mathbb{Z}_p})$. The source of $\Phi_{\mathbb{Z}_p}$ commutes with pullback, as in the proof of Theorem 6.1. By Theorem 6.2, $j^*\Phi_{\mathbb{Z}_p}$ and $i^*\Phi_{\mathbb{Z}_p}$ are MGL-module maps to $H\mathbb{Z}$ extending the unit, so by the uniqueness of Lemma 1.4 over a field they are $\Phi_{\mathbb{Q}_p}$ and $\Phi_{\mathbb{F}_p}$. Hence $j^*C = 0$ by the theorem in characteristic zero [H, Theorem 7.12], and $i^*C = 0$ by Theorem 6.1. The localization cofibre sequence $j_!j^*C \to C \to i_*i^*C$ in $\mathrm{SH}(\operatorname{Spec}\mathbb{Z}_p)$ now shows that $C = 0$. □

Theorems 6.1 and 6.3 are the two parts of Theorem A. Corollary B follows, since an equivalence remains an equivalence after applying $\operatorname{cofib}(p^r\colon - \to -)$ for every r and then passing to the limit over r.

7. Consequences for additive theories

Theorem 6.1 identifies the quotient of MGL by the coefficients of the universal formal group law with motivic cohomology. As a consequence, every MGL-algebra with additive formal group law receives its orientation through $H\mathbb{Z}$. Since the class of a finite syntomic morphism in $H\mathbb{Z}^{0,0}$ is its degree, such a theory sees only the degree of a finite syntomic morphism.

Proposition 7.1. *Let* k *be a field of characteristic* p *and* E *a commutative* MGL*-algebra in* $\mathrm{SH}(k)$ *with additive formal group law. The structure map* $\mathrm{MGL} \to E$ *factors through* $H\mathbb{Z}$ *as a map of* MGL*-modules. Hence, for a smooth* k*-scheme* X *and finite syntomic* $f\colon W \to X$ *of degree* d*, we have* $f_*^E 1 = d$ *in* $E^{0,0}(X)$.

Proof. We extend the structure map over the cofibres of the a_i one at a time. Since the formal group law of E is additive, the classifying map $L \to \pi_{2*,*}E$ kills every a_i. Suppose given an MGL-module map $g_N\colon \mathrm{MGL}/(a_1, \ldots, a_N) \to E$ extending the structure map. Because g_N is MGL-linear, the composite $g_N \circ a_{N+1}$ equals $a_{N+1} \cdot \Sigma^{2N+2,N+1} g_N$. This composite is null, because a_{N+1} acts on E through its image in $\pi_{2N+2,N+1}E$, which is 0. Hence g_N extends to $\mathrm{MGL}/(a_1, \ldots, a_{N+1})$. Passing to the colimit over N, we obtain a unit-preserving MGL-module map $\mathrm{MGL}/(a_1, a_2, \ldots) \to E$. By Theorem 6.1, the map Φ_k identifies its source with $H\mathbb{Z}$ as an MGL-module, which gives the factorization.

For the second assertion, $f_*^E 1$ is the image of $f_*^{\mathrm{MGL}} 1$ under the structure map. The image of $f_*^{\mathrm{MGL}} 1$ in $H\mathbb{Z}^{0,0}(X) = \mathbb{Z}^{\pi_0(X)}$ is the rank d of f, and a unit-preserving MGL-module map sends $d \in H\mathbb{Z}^{0,0}(X)$ to $d \cdot 1 \in E^{0,0}(X)$. □

Corollary 7.2. *In the situation of Theorem 4.1, for every commutative* p*-local* MGL*-algebra* E *with additive formal group law, the necklace class* b *maps to* $\frac{d^p - d}{p}$ *in* $E^{0,0}(X)$.

Proof. By Proposition 7.1, the map $\mathrm{MGL}_{(p)} \to E$ factors as the orientation $\mathrm{MGL}_{(p)} \to H\mathbb{Z}_{(p)}$ followed by a unit-preserving MGL-module map $H\mathbb{Z}_{(p)} \to E$. As in the proof of that proposition, the second map sends $c \in H\mathbb{Z}^{0,0}_{(p)}(X) = \mathbb{Z}^{\pi_0(X)}_{(p)}$ to $c \cdot 1$, so it suffices to compute the image of b in $H\mathbb{Z}^{0,0}_{(p)}(X)$. The orientation $\mathrm{MGL}_{(p)} \to H\mathbb{Z}_{(p)}$ equals $\Phi \circ \rho$, since both are $\mathrm{MGL}_{(p)}$-module maps sending 1 to 1. It is a ring map and sends u to d by Proposition 7.1. By (4.1), $u^p - u - pb \in \ker \rho$, and applying $\Phi \circ \rho$ gives

$$d^p - d = p\,\Phi(\rho(b)) \quad \text{in } H\mathbb{Z}^{0,0}_{(p)}(X) = \mathbb{Z}^{\pi_0(X)}_{(p)}.$$

This group is torsion free, so the image of b is $\frac{d^p - d}{p}$, whichever lift b was chosen. □

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